

Chapter 1 Fundamental Groups

1.1 Definitions

Definition 1.1.0.1 Let X be a topological space:

- We define a **path** in X as a continuous map

$$\gamma : [a, b] \rightarrow X$$

often we say: $\gamma(a)$ is the **starting point** and $\gamma(b)$ is the **end point**.

- A **loop** is a path with $\gamma(a) = \gamma(b)$;
- Given a path $\gamma : [a, b] \rightarrow X$, define its inverse

$$\bar{\gamma} : [a, b] \rightarrow X \quad t \mapsto \gamma(b + a - t)$$

1.1.1 Path-Homotopy

Definition 1.1.1.1 Let X be a topological space:

- Let $\gamma_0, \gamma_1 : [a, b] \rightarrow X$ be two paths with the same starting point P , and the same end point Q .

A **path homotopy** between the paths γ_0, γ_1 is a homotopy rel $\{a, b\}$ between γ_0 and γ_1 .

In other words, a path-homotopy is a continuous map

$$\Gamma : [a, b] \times [0, 1] \rightarrow X$$

Such that

$$\begin{aligned} \Gamma(t, 0) &= \gamma_0(t) & \Gamma(t, 1) &= \gamma_1(t) \\ \Gamma(a, s) &= P & \Gamma(b, s) &= Q \end{aligned}$$

- If γ_0 and γ_1 are path-homotopic, then written as $\gamma_0 \simeq_p \gamma_1$;
- Especially, a loop γ is **null-homotopic**, if it is path-homotopic to the constant path δ given by

$$\delta(t) := \gamma(a)$$

Reparametrization

Definition 1.1.1.2 Let $\gamma : [a, b] \rightarrow X$ be a path in topological space X , and let

$$\varphi : [a, b] \rightarrow [a, b]$$

which satisfies:

- Continuous;
- $\gamma(a) = a; \gamma(b) = b$;
- Monotone;

◇ **Lemma 1.1.1.3** (Reparametrization-Homotopy Lemma) Let $\gamma : [a, b] \rightarrow X$ be a path in a topological space X .

If: a path $\delta : [a, b] \rightarrow X$ is obtained from γ by a reparametrization, then

$$\gamma \simeq_p \delta$$

◦ **Proof** Firstly, consider the reparametrization $\gamma : [a, b] \rightarrow [a, b]$. Then

$$\begin{aligned} F &: [a, b] \times [0, 1] \rightarrow X \\ (t, s) &\mapsto \gamma(t + (1 - s)\varphi(t)) \end{aligned}$$

is a path homotopy. □

◇ **Lemma 1.1.1.4** (Path-Homotopy Equivalence Lemma) Let X be topological space and $x, y \in X$. Then, path-homotopy is an equivalence relation on the set of paths $[a, b] \rightarrow X$.

◦ **Proof** ... □

Homotopy Class

◇ **Lemma 1.1.1.5** ...

◦ **Proof** ... □

Definition 1.1.1.6

- (Homotopy Class)
- (Product)

♠ **Proposition 1.1.1.7** (Path-Homotopy Product Proposition)

The Fundamental Group of Pointed Topological Space

Definition 1.1.1.8 ...

♠ **Proposition 1.1.1.9** (π_1 -Group Proposition) Let X be a topological space and $x_0 \in X$, the set

$$\pi_1(X, x_0) := \{\text{path-homotopy classes of loops in } (X, x_0)\}$$

together with

◦ **Proof** ... □

Definition 1.1.1.10 ...

Change of Base Points

♠ Proposition 1.1.1.11 (Change of Base Point Proposition)

◦ Proof ...

□

★ Corollary 1.1.1.12 ...

◦ Proof ...

□

Examples

- Fundamental Group of the circle:

◊ **Lemma 1.1.1.13** (Lebesgue's Lemma) Let

$$f : [0, 1]^m \rightarrow X$$

be a continuous map. Let $\{V_i\}$ be an open cover of X .

Then, for every $a_1, \dots, a_m \in \{0, 1, \dots, N-1\}$ there exists an $N \in \mathbb{N}$ such that

$$f\left(\left[\frac{a_1}{N}, \frac{a_1+1}{N}\right] \times \dots \times \left[\frac{a_m}{N}, \frac{a_m+1}{N}\right]\right) \subseteq V_i$$

◦ Proof Here:

□

Definition 1.1.1.14

- Consider map $\Xi : \mathbb{R} \rightarrow S^1$;
- Let X be a topological space with $f : X \rightarrow S^1$ continuous.
A **lift** of f is a continuous map

$$\tilde{f} : W \rightarrow \mathbb{R}$$

such that $\Xi \circ \tilde{f} = f$. In other words, such that for all $w \in W$ we have

$$f(w) = \exp(i \cdot \tilde{f}(w))$$

$$\begin{array}{ccc} & & \mathbb{R} \\ & \nearrow \tilde{f} & \downarrow p \\ W & \xrightarrow{f} & S^1 \end{array}$$

♣ **Example 1.1.1.15** (Lift) Consider the map $f : [0, 1] \rightarrow S^1, t \mapsto \exp(2\pi i n t)$. Then, consider the lift given by

$$\tilde{f} : t \mapsto 2\pi n \cdot t$$

The lift is not unique, take $k = 2\pi \cdot n$.

◊ **Lemma 1.1.1.16** (Lift Uniqueness Lemma) Let W be a connected topological space. Let $f : W \rightarrow S^1$ be a continuous map and $\tilde{\varphi}, \tilde{\psi} : W \rightarrow \mathbb{R}$ be two lifts of f .

If $\tilde{\varphi}, \tilde{\psi}$ agrees on some point $w_0 \in W$, then they agree everywhere.

◦ **Proof** Consider the following map

$$\theta : W \rightarrow \mathbb{R} \quad w \mapsto \theta(w) := \tilde{\varphi}(w) - \tilde{\psi}(w)$$

Then, by hypothesis, we have

$$\theta(w_0) = 0 \quad \exp(i\theta(w)) = f(w) \cdot f(w)^{-1} = 1$$

This implies, that $s \mapsto \theta(w)$ maps W into $2\pi \cdot \mathbb{Z} = \Xi^{-1}(1)$. W is connected, so they agree everywhere. $\square$

◊ **Lemma 1.1.1.17** (Lift-Extension Lemma) Let X be a topological space, we suppose that we have a decomposition that

$$X = A \cup B \quad A, B \text{ are closed}$$

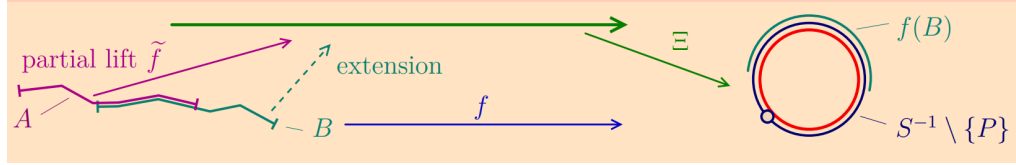
Now, let $f : X \rightarrow S^1$ be continuous, $\tilde{f} : A \rightarrow \mathbb{R}$ be a lift of $f|_A$. We suppose that the following statements hold:

- There exists a $P \in S^1$ with

$$f(B) \subseteq S^1 - \{P\}$$

- The intersection $A \cap B$ is **connected** and **nonempty**.

Then, there exists a lift $\tilde{f} : X \rightarrow \mathbb{R}$ of f agrees with the given lift on A .



◦ **Proof** Firstly, pick some $x_0 \in A \cap B$, and $s \in \mathbb{R}$ such that $\Xi(s) = P$. Then,

$$f(x_0) \in f(A \cap B) \subseteq f(B) \subseteq S^1 - \{P\} \Rightarrow \tilde{f}(x_0) \notin \Xi^{-1}(P)$$

So, there exists some $m \in \mathbb{Z}$ such that

$$\tilde{f}(x_0) \in (s + 2\pi m, s + 2\pi(m + 1))$$

Then, restriction of Ξ to $(s + 2\pi m, s + 2\pi(m + 1)) \rightarrow S^1 - \{P\}$ is a homeomorphism. Denote $\Psi := [\Xi|(s + 2\pi m, s + 2\pi(m + 1))]^{-1}$.

Now, define

$$\tilde{f}(x) := \begin{cases} \tilde{f}(x) & x \in A \\ \Psi(f(x)) & x \in B \Rightarrow f(x) \in S^1 - \{P\} \end{cases}$$

The definition agrees on $A \cap B$: For $x \in A \cap B$,

$$\Psi(f(x)) = \tilde{f}(x), x \in A$$

Thus, the definition of $\tilde{f}$ can be extended to $A \cup B$. $\square$

◇ **Lemma 1.1.1.18** (Lift from S^1 to $\mathbb{R}$) Given any continuous map

$$f : [0, 1] \rightarrow S^1$$

and any $\tilde{y}_0 \in \mathbb{R}$ with

$$f(0) = \exp(i \cdot \tilde{y}_0) = \Xi(\tilde{y}_0)$$

There exists a **unique lift** $\tilde{f} : [0, 1] \rightarrow \mathbb{R}$ of f with

$$\tilde{f}(0) = \tilde{y}_0$$

◦ **Proof**

- Uniqueness: By Lift Uniqueness lemma.
- Existence: Apply Lift-Extension lemma. Decompose the interval $[0, 1]$ into smaller intervals, such that: on each subinterval, the image of f misses at least one point in S^1 .

□

► **Theorem** The map

$$\Psi : \mathbb{Z} \rightarrow \pi_1(S^1, 1)$$

$$\Theta : \pi_1(S^1, z_0) \rightarrow \mathbb{Z}$$

are well defined, and are isomorphisms of group.

◦ **Proof**

□

• Lifting of Square:

◇ **Lemma 1.1.1.19** ...

◦ **Proof** ...

□

• Fundamental Group of spheres:

◇ **Lemma 1.1.1.20** (Sphere Minus Point Lemma) For every $P \in S^n$, $S^n - \{P\}$ is simply connected.

◦ **Proof** We have $S^n - \{P\} \cong \mathbb{R}^n$.

□

♠ **Proposition 1.1.1.21** For $n \geq 2$ we have

$$\pi_1(S^n) = 0$$

i.e. it is simply connected.

◦ Proof ...

□

♠ **Proposition 1.1.1.22** (π_1 -Group Proposition) Let X be a topological space and $x_0 \in X$, the set

$$\pi_1(X, x_0) :=$$

together with product

◦ Proof ...

□

Definition 1.1.1.23 ...

Alternative Descriptions

Change of Base Points

1.1.2 Applications

The Fundamental Theorem of Algebra

► **Theorem** (TFTA) Every nonconstant polynomial in $\mathbb{C}[z]$ has a zero in $\mathbb{C}$.

◦ Proof ...

□

2D-Borsuk-Ulam Theorem

Path Integrals, Fundamental Groups

1.1.3 Functoriality

Group Homomorphisms induced by Continuous Map

Definition 1.1.3.1 ...

1.2 Coverings and Fundamental Groups

1.2.1 Group Action

Klein Bottle as Quotient

◇ **Lemma 1.2.1.1** ...

◦ Proof ...

□

1.2.2 Covering of Topological Spaces

Definition 1.2.2.1 Let $p : X \rightarrow B$ be a continuous map between topological spaces:

- An open subset $U \subseteq B$ is called **uniformly covered** if
- We say that the map $p : X \rightarrow B$ is a covering, if p is surjective and if for every $b \in B$, there exists an open neighborhood U of b which is uniformly covered;
- Sometimes X is called a **covering space** B .

◇ **Lemma 1.2.2.2** (Covering-Preimage Lemma) Let $p : X \rightarrow B$ be a covering of topological spaces. If B is a connected topological space, then for any two point $P, Q \in B$ we have

$$\#p^{-1}(P) = \#p^{-1}(Q)$$

◦ **Proof ...**

□

Definition 1.2.2.3

-
-
-

Action-Covering Proposition

♠ **Proposition 1.2.2.4** ...

◦ **Proof ...**

□

1.2.3 Basic Properties of Coverings

♠ **Proposition 1.2.3.1** Let $p : X \rightarrow Y$ be a covering.

- Let γ be a path in Y with $p(x) = \gamma(0)$. Then there exists a unique $\dot{\gamma}$ in X such that

$$p \circ \dot{\gamma} = \gamma$$

- Let H be the homotopy between the paths γ_0, γ_1 , in Y and $l_1 : [0, 1] \rightarrow X$, be such that

$$p(l_1(t)) = H(0, t)$$

then, there exists unique $\dot{H} : [0, 1] \times [0, 1] \rightarrow X$ such that

$$\dot{H}(0, t) = l_1(t), p(\dot{H})$$

★ **Corollary 1.2.3.2** Let $p : (X, x) \rightarrow (Y, y)$ covering of pointed spaces, then

$$p_* : \pi_1(X, x) \rightarrow \pi_1(Y, y)$$

is injective.

★ **Corollary 1.2.3.3** Let γ be a loop in (Y, y) and $\dot{\gamma}$ a lift of γ starting at x , then $\dot{\gamma}(1) \in p^{-1}(x)$ depends only on the homotopy class of γ .

This defines a map

$$\pi_1(Y, y) \rightarrow p^{-1}(y)$$

If x in $p_*\pi_1(Y, y) \subseteq p^{-1}(x)$, $x(1) = x$,

$$\pi_1(X, y)/p_*\pi_1(X, x) \rightarrow p^{-1}(y)$$

◦ **Proof ...**

□

♠ **Proposition 1.2.3.4** (Classification of Covering Space) Assume (Y, y) connected, locally path connected, semilocally simply connected. One assignment

$$(X, x) \xrightarrow{p} (Y, y)$$

★ **Corollary 1.2.3.5** Any space (Y, y) as above has a simply connected covering space, unique up to iso

Definition 1.2.3.6 Let $p: (X, x) \rightarrow (Y, y)$ its deck transformation group $Aut(p)$ is the group of automorphisms of p ;

♠ **Proposition 1.2.3.7** Let $N \subseteq Aut(p)$ be the character of $p_*\pi_1(X, x)$ then $Aut(p) \rightarrow p^{-1}(y)$, $\varphi \mapsto \varphi(x)$, $\pi_1(Y, y)/p_*\pi_1(X, x)$ have the same image in $p^{-1}(y)$. And composition is an iso of groups.

Definition 1.2.3.8 p is normal if $N = \pi_1(x, y)$.

★ **Corollary 1.2.3.9** Let p be normal, then:

- $Aut(p) = \pi_1(Y, y)/\pi_1(X, x)$;
- The action of $Aut(p)$ on $p^{-1}(y)$ is transitive;
- p induces a homeomorphism

$$X/Aut(p) \cong Y$$

◦ **Proof ...**

□

★ **Corollary 1.2.3.10** Let $p: (X, x) \rightarrow (Y, y)$ be universal covering, let γ_0, γ_1 loops in (Y, y) , TFAE:

- $X_0 \sim X_1$;
- $\gamma_0(1) = \gamma_1(1)$;

Moreover, $\gamma \leftrightarrow \varphi$

1.2.4 Lifts

1.2.5 Pullbacks

1.3 Group Actions, and Fundamental Groups

1.4 Applied Topology

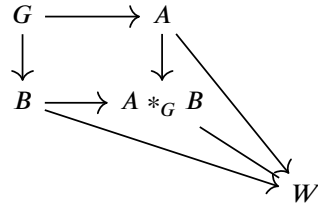
1.5 Covering of Smooth and Topological Manifolds

1.6 title

1.7 Seifert van Kampen Theorem

Firstly, the notion

$$A *_G B := (A * B) / \langle \langle \{\alpha(g) * \beta(g)^{-1}\}_{g \in G} \rangle \rangle, \alpha : G \rightarrow A, \beta : G \rightarrow B$$



◇ Lemma 1.7.0.1

- The group homomorphism $\Xi : \pi_1(U, x_0) * \pi_1(V, x_0)$ is surjective for: Topological space X and $U, V \subseteq X$ be open subsets such that
 - $X = U \cup V$;
 - $U \cap V$ is path-connected and non-empty;

We choose a base point $x_0 \in U \cap V$, then $x_0 \in U \cap V$.

- Every loop $s : [0, 1] \rightarrow X$ in x_0 is path-homotopic to a loop which is the concatenation of finitely many loops, each of which lies in (U, x_0) or in (V, x_0) .

◦ **Proof** Let $s : [0, 1] \rightarrow X = U \cup V$ be a loop. Then, U, V are open, apply Lebesgue number lemma: There exists a subdivision

$$0 = t_0 < t_1 < \dots < t_{k+1} = 1$$

of the interval $[0, 1]$ such that the image each $s_i = s|_{[t_i, t_{i+1}]}$ lies in U or in V .

Then, by taking union of these intervals, we can satisfy that

$$s(t_i) \in U \cap V$$

But such s_i may be not a loop. We could construct loop for each s_i by following steps: Notice that $U \cap V$ is path-connected. So we could take another path $s(t_{i+1}) \rightarrow s_i$ in $U \cap V$. So we have

$$s \simeq_p s_0 * \dots * s_k \simeq_p (s_0 * p_1) * (\bar{p}_1 * s_1 * p_2) * \dots$$

□

► **Theorem** (Seifert-van-Kampen Theorem) Let X be a topological space and $X = U \cup V$ be a decomposition of X in two open subsets U and V such that: $U \cap V$ is

- nonempty;
- path-connected

Let $x_0 \in U \cap V$. We denote $i : U \cap V \rightarrow U$ and $j : U \cap V \rightarrow V$ the inclusions:

- The inclusions $k : U \rightarrow X$ and $l : V \rightarrow X$ induce an isomorphism

$$\Phi : \pi_1(U, x_0) * \pi_1(V, x_0) \xrightarrow{\cong} \pi_1(X, x_0)$$

which has the property that the following diagram commutes

$$\begin{array}{ccccc}
 \pi_1(U \cap V, x_0) & \xrightarrow{i_*} & \pi_1(U, x_0) & & \\
 \downarrow j_* & & \downarrow & \searrow k_* & \\
 \pi_1(V, x_0) & \xrightarrow{\quad} & \pi_1(U, x_0) *_{\pi_1(U \cap V, x_0)} \pi_1(V, x_0) & \xrightarrow{\Phi} & \pi_1(X, x_0) \\
 & \searrow l_* & & \nearrow \cong & \\
 & & & &
 \end{array}$$

group $\pi(X, x_0)$ is a pushout.

- Especially: If $U \cap V$ is simply connected, then the inclusions $U \rightarrow V$ and $V \rightarrow X$ induce an isomorphism

$$\Phi : \pi_1(U, x_0) * \pi_1(V, x_0) \rightarrow \pi_1(X, x_0)$$

◦ **Proof** Notice that: There exists pushout $\Phi : \pi_1(U, x_0) * \pi_1(V, x_0) \rightarrow \pi_1(X, x_0)$:

- Surjective: Consider following diagram

$$\begin{array}{ccc}
 \pi_1(U, x_0) * \pi_1(V, x_0) & & \\
 \downarrow \Xi & \searrow & \\
 \pi_1(U, x_0) *_{\pi_1(U \cap V, x_0)} \pi_1(V, x_0) & \longrightarrow & \pi_1(X, x_0)
 \end{array}$$

Ξ given by the products. So surjective.

- Injective: $\Phi : \pi_1(U, x_0) *_{\pi_1(U \cap V, x_0)} \pi_1(V, x_0) \rightarrow \pi_1(X, x_0)$ which is induced by the inclusion $i : U \rightarrow X$ and $j : V \rightarrow X$ is injective.

We just need to show: Exists map $\Psi : \pi_1(X, x_0) \rightarrow \pi_1(U, x_0) *_{\pi_1(U \cap V, x_0)} \pi_1(V, x_0)$ such that

$$\Psi \circ \Phi = id$$

□

1.7.1 Suspension

Definition 1.7.1.1

◊ **Lemma 1.7.1.2** (Suspension- π_1 -Lemma)

1.7.2 Fundamental Groups of Wedeges

Definition 1.7.2.1 (Good Point)

◇ **Lemma 1.7.2.2** ...

◦ **Proof** ...

□

1.8 Classification of 2-dimensional Manifold

1.9 Decision Problem

1.10 Directed Colimits

1.11 HNN-Seifert-Van Theorem

1.12 Curves and Surfaces

1.13 Introduction to Higher Homotopy Theory

Chapter 2 The Universal Covering

2.1 The Existence of Universal Covering

Definition 2.1.0.1 Let $p : X \rightarrow B$ continuous map between topological space X, B

- (Uniformly Covered) An open subset $U \subseteq B$ is called **uniformly covered**, if

$$p^{-1}(U) = \sqcup_{i \in I} V_i$$

with the property:

$$p|_{V_i} : V_i \rightarrow U$$

is a homeomorphism .

- (Covering) We say that the map

$$p : X \rightarrow B$$

is a **covering**, if:

- p is surjective;
- For every $b \in B$, there exists an open neighborhood of b which is uniformly covered.
- (Covering of Pointed Spaces) A covering $p : (X, x_0) \rightarrow (B, b_0)$ of pointed spaces, if $p : X \rightarrow B$ is a covering and with

$$p(x_0) = b_0$$

- (Path-Connected Covering) A covering $p : X \rightarrow B$ is **path-connected**, if X is path-connected.

Question: Does every path-connected non-empty topological space B admit a path-connected covering $p : X \rightarrow B$ such that X is simply connected?

2.1.1 Local Properties

Definition 2.1.1.1 Let P be a property. We say that: A topological space is **locally** P if given any $x \in X$, and any neighborhood U of x , there exists an open neighborhood V contained in U such that V has the property P .

2.1.2 Lifting Maps to Coverings

Definition 2.1.2.1 Let $p : (X, x_0) \rightarrow (B, b_0)$ be a covering of pointed topological spaces.

Furthermore, let $(Z, z_0) \rightarrow (B, b_0)$ be a continuous map, a **lift** is a continuous map $\tilde{f} : (Z, z_0) \rightarrow (X, x_0)$ which make the following diagram commutes:

$$\begin{array}{ccc} & & (X, x_0) \\ & \nearrow \tilde{p} & \downarrow p \\ (Z, z_0) & \longrightarrow & (B, b_0) \end{array}$$

Uniqueness

♠ **Proposition 2.1.2.2** (Lift Uniqueness Proposition) Let $X \rightarrow B$ be a covering and let Z be a connected topological space.

Furthermore, let $f : Z \rightarrow B$ with liftings $\hat{f}, \tilde{f} : Z \rightarrow X$. Then: if $\hat{f}, \tilde{f}$ agrees at some $z_0 \in Z$, then they agree everywhere.

◦ **Proof Here:**

□

Path-Lifting

♠ **Proposition 2.1.2.3** ...

◦ **Proof ...**

□

Loop-Lifting

♠ **Proposition 2.1.2.4** ...

◦ **Proof ...**

□

Maps-Lifting

♠ **Proposition 2.1.2.5** Let $p : (X, x_0) \rightarrow (B, b_0)$ be a covering of pointed topological spaces.

Furthermore, let Z be a **path-connected** topological space, and $f : (Z, z_0) \rightarrow (B, b_0)$ be a continuous map.

If Z is locally path-connected, then we have following equivalent statements:

$$\begin{array}{ccc} & (X, x_0) & \\ \nearrow \exists \tilde{f} & \downarrow p & \\ (Z, z_0) & \xrightarrow{f} & (B, b_0) \end{array} \Leftrightarrow im(f_*) \subseteq im(p_*)$$

◦ **Proof ...**

□

Concatenation of Path Lifting

♠ **Proposition 2.1.2.6** Let $p : (X, x_0) \rightarrow (B, b_0)$ be a covering of pointed topological spaces, and γ be a path in B starting point b_0 .

◦ **Proof ...**

□

2.1.3 Existence of Covering Spaces**2.2 Deck Transformation Group**

Chapter 3 Knot and Links

3.1 Introduction

3.1.1 Knot and Links

◇ **Lemma 3.1.1.1** (Stereographic Projection Lemma) We refer to the map

$$\Phi : S^3 \rightarrow \mathbb{R}^3 \cup \{\infty\}$$

$$(x_1, x_2, x_3, x_4) \mapsto \begin{cases} (\frac{x_1}{1-x_4}, \frac{x_2}{1-x_4}, \frac{x_3}{1-x_4}) & x_4 < 1 \\ \infty & x_4 = 1 \end{cases}$$

- The map Φ sends the north pole $N = (0, 0, 0, 1)$ to ∞ ;
- For any $P \in S^3$, that does not equal the North Pole N , the point $\Phi(P) \in \mathbb{R}^3$ is the unique point such that the ray emanating
-
- Φ is a homeomorphism;
- The restriction

◦ **Proof ...**

□

Remark S^3 can be seen as

$$S^3 =_i \{(z, w) \in \mathbb{C}^2 \mid |z|^2 + |w|^2 = 1\}$$

Under this homeomorphism, $\mathbb{R}^3$ should be seen as a submanifold of S^3 .

Definition 3.1.1.2

- A **knot** is a submanifold of S^3 which is diffeomorphic to S^1 .
- Especially:

3.1.2 Orientations

Chapter 4 Simplicial Complexes

4.1 Definitions

Definition 4.1.0.1 An **abstract simplicial complex** is a pair (V, S) where V is a set and S is a subset of the power set $P(V)$ of V such that the following conditions are satisfied:

- Each $s \in S$ is finite and nonempty;
- If $s \in S$, then any non-empty subset of s also lies in S ;
- Given any $v \in V$, the set $\{v\} \in S$.

We refer to the elements of V the **vertices of the abstract simplicial complex** and we refer to the elements of S as the **simplices of the abstract simplicial complex**.

4.1.1 Dimension

Especially:

Definition 4.1.1.1 Let $V = (K, S)$ be an abstract simplicial complex:

- We say $K = (V, S)$ is **finite** if V is finite. Similarly we say that $K = (V, S)$ is **countable** if V is countable.
- If $s \in S$ has $k + 1$ elements, then s is called a **k -dimensional simplex**, or shorter a **k -simplex** of (V, S) ;
- If $K = (V, S)$ is the empty abstract simplicial complex, then we define its dimension as -1 .

Now suppose that $K = (V, S)$, then

$$\dim(K) := \max\{\text{card}(s) - 1 \mid s \in S\}$$

4.1.2 Face

Definition 4.1.2.1 Let $K = (V, S)$ be an abstract simplicial complex, let t be a simplex of K , a **face** of t is a simplex $s \in S$ with $s \subseteq t$.

◇ **Lemma 4.1.2.2** (Simplicial-Intersection Lemma) Let $K = (V, S)$ be an abstract simplicial complex. Furthermore, let $s, t \in S$. Then, either the intersection $s \cap t = \emptyset$ or $s \cap t$ is again a simplex of K and it is a face of s and t .

◦ **Proof Omitted.**

□

4.1.3 Subcomplexes

Definition 4.1.3.1 Let $L = (W, T)$ be an abstract simplicial complex. An **abstract subcomplex** of $L = (W, T)$ is a pair $K = (V, S)$ such that $V \subseteq W, S \subseteq T$ with $K = (V, S)$ is a simplicial complex.

◇ **Lemma 4.1.3.2** Let $\{L_i = (W_i, T_i)\}_{i \in I}$ be a family of abstract simplicial complexes, the following statements hold:

- $\cap L_i := (\cap W_i, \cap T_i), \cup L_i := (\cup W_i, \cup T_i)$ are abstract simplicial complexes;
- $\cap L_i$ is a subcomplex of each L_i , and each L_i is subcomplex of $\cup L_i$;

- If each L_i is subcomplex of K , so is $\cup L_i$. i.e. $\cup L_i$ is the smallest complex containing L_i .

◦ **Proof ...**

□

4.1.4 Simplicial Map

Definition 4.1.4.1 Let $K = (V, S)$ and $L = (W, T)$ be abstract simplicial complexes:

- A **simplicial map** $f : K \rightarrow L$ is defined as a map $f : V \rightarrow W$ such that for $s \in S$, we have $f(s) \in T$.
i.e. A simplicial map induces a map $S \rightarrow T$.
- A simplicial isomorphism is a simplicial map that is a bijection on the set of vertices on the set of simplices.

4.1.5 Category of Simplicial Complexes

Definition 4.1.5.1 ...

4.1.6 Topological Relization

4.2 Simplicial Approxiamation Theorem

4.2.1 SAT

4.3 Simplicial Homology

Definition 4.3.0.1 Let $K = (V, S)$ be an ordered abstract simplicial complex: Given $n \in \mathbb{N}_0$, define

$$C_n^{simp, \leq}(K) := \mathbb{Z}[S_n]$$

where S_n is the set of n -simplices of K . Furthermore, we consider the boundary map

$$\begin{aligned} \partial : C_n^{simp, \leq}(K) &\rightarrow C_{n-1}^{simp, \leq}(K) \\ \{v_0 < \dots < v_n\} &\mapsto \sum_{i=0}^n (-1)^i \cdot \{v_0, \dots, v_{i-1}, \hat{v}_i, v_{i+1}, \dots, v_n\} \end{aligned}$$

◇ **Lemma 4.3.0.2** (Simplicial Boundary Lemma) Let $K = (V, S)$ be an ordered abstract simplicial complex. Given any $n \in \mathbb{N}_0$ we have

$$\partial_{n-1} \circ \partial_n = 0$$

In other word: $(C_*^{simp, \leq}(K), \partial)$ is a chain complex.

◦ **Proof ...**

□

Definition 4.3.0.3 ...

4.3.1

◇ Lemma 4.3.1.1 ...

◦ Proof ...

□

◇ Lemma 4.3.1.2 ...

◦ Proof ...

□

4.3.2 Simplicial Homology

4.4 Simplicial and Singular Homology

4.5 The Nerve Theorem

Definition 4.5.0.1 Let X be a topological space and $\mathcal{U} := \{U_i\}_{i \in I}$ be a cover of X :

-
-

4.5.1 Nerve

Let X be a set, $\mathcal{U} = \{U_i\}_{i \in I}$ be a family of subsets of X . We refer to the abstract simplicial complex that is given by: The vertex set I and the simplicial set

...

4.6 Simplicial Structures

4.7 Simplicial Sets

4.7.1 Definitions

Definition 4.7.1.1

- Let Δ be the category of finite ordinal numbers, with order-preversing maps between them. More precisely, the objects for Δ consist of elements $n, n \geq 0$, where n is a string of relations

$$0 \rightarrow 1 \rightarrow \cdots \rightarrow n$$

- Morphisms

$$\theta : m \rightarrow n$$

where θ is (strictly) order-preserving set function.

Definition 4.7.1.2

- A **simplicial set** is a covariant functor

$$X : \Delta^{op} \rightarrow Sets$$

where $Sets$ is the category of sets.

- A map $f : X \rightarrow Y$ of simplicial sets is a natural transformation of contravariant set-valued functors defined on Δ

$$\Delta^{op} \begin{array}{c} \xrightarrow{X} \\ \Downarrow f \\ \xrightarrow{Y} \end{array} Sets$$

- Take the following notation S for simplicial sets.

♣ **Example 4.7.1.3** (Topological Standard n -Simplex) Among the functors $sm \rightarrow n$ appearing in Δ there are special ones, namely

Definition 4.7.1.4 Firstly, write down the sets

$$Y_n := Y(n)$$

Define the morphisms in Δ

- $d^i : n-1 \rightarrow n, 0 \leq i \leq n, (0 \rightarrow \dots \rightarrow n-1) \mapsto (0 \rightarrow \dots \rightarrow i-1 \rightarrow i+1 \rightarrow \dots \rightarrow n)$
- $s^j : n+1 \rightarrow n, 0 \leq j \leq n, (0 \rightarrow 1 \rightarrow \dots \rightarrow n+1) \mapsto (0 \rightarrow \dots \rightarrow j \xrightarrow{1} j \rightarrow \dots \rightarrow n)$

We have:

♠ **Proposition 4.7.1.5** (Cosimplicial Identities) We have

$$\begin{aligned} d^j d^i &= d^i d^{j-1}, i < j \\ s^j d^i &= \begin{cases} d^i s^{j-1} & i < j \\ 1 & i = j \\ 1 & i = j+1 \\ d^{i-1} s^j & i > j+1 \end{cases} \\ s^j s^i &= s^i s^{j+1}, i \leq j \end{aligned}$$

◦ **Proof ...**

□

Then, the relation on simplicial sets

Definition 4.7.1.6 For a simplicial set $Y : \Delta^{op} \rightarrow Sets$, define

$$\begin{aligned} d_i &: Y_n \rightarrow Y_{n-1} \\ s_j &: Y_n \rightarrow Y_{n+1} \end{aligned}$$

♠ **Proposition 4.7.1.7** (Simplicial Identities) We have

$$\begin{aligned} d_i d_j &= d_{j-1} d_i, i < j \\ d_i s_j &= \begin{cases} s_{j-1} d_i & i < j \\ 1 & i = j \\ s_j d_{i_1} & i > j \end{cases} \\ s_i s_j &= s_{j+1} s_i, i \leq j \end{aligned}$$

◦ **Proof ...**

□

Simplicial Abelian Group

Definition 4.7.1.8 Let Y be a simplicial set. Then, define the simplicial abelian group, with a chain complex called **Moore complex** written $\mathbb{Z}Y$, with

$$\cdots \rightarrow \mathbb{Z}Y_2 \rightarrow \mathbb{Z}Y_1 \rightarrow \mathbb{Z}Y_0$$

With differential map

$$\partial = \sum_{i=0}^n (-1)^i d_i$$

in degree n .

Classifying Space

Definition 4.7.1.9 Suppose that C be a small category. The classifying space BC of C is the simplicial set with

$$BC_n = \text{hom}_{\text{cat}}(n, C)$$

In other word

$$\begin{array}{ccccccc} 0 & \longrightarrow & \cdots & \longrightarrow & i & \longrightarrow & \cdots & \longrightarrow & n \\ \downarrow & & & & \downarrow & & & & \downarrow \\ a_0 & \longrightarrow & \cdots & \longrightarrow & a_i & \longrightarrow & \cdots & \longrightarrow & a_n \end{array}$$

This gives a homotopy theoretic point of view.

Standard n-simplex, Simplicial Δ^n

Definition 4.7.1.10 Define the contravariant functor Δ :

$$\Delta^n := \text{hom}_{\Delta}(-, n) \in S$$

here, S is the category of simplicial sets.

With Yoneda's lemma: For simplicial set $Y : \Delta^{op} \rightarrow \text{Sets}$ we have bijection

$$\text{Hom}_S(\Delta^n, Y) \cong Y_n$$

When Y_n is collection of n -simplices of Y . Precisely(Construction by Yoneda): Firstly, write $\iota_n = 1_n \in \text{hom}_\Delta(n, n) \Rightarrow \varphi(\iota_n) \in Y_n$. Given a simplicial map $\varphi : \Delta^n \rightarrow Y$, we have $\varphi(\iota_n)$. Then, the correspondence

$$\varphi \Leftrightarrow \varphi_n(\iota_n)$$

For $x \in Y_n$, it associate to a unique simplicial map

$$\delta_x : \Delta^n \rightarrow Y$$

such that $\iota_x(\iota_n) = x$.

Boundaries,Korn

Definition 4.7.1.11

- The simplicial set $\partial\Delta^n$ is the smallest subcomplex of Δ^n containing the faces

$$d_j(\iota_n), 0 \leq j \leq n$$

of the standard simplex ι_n . Specified in j -simplices by

$$\partial\Delta_j^n := \begin{cases} \Delta_j^n \\ \text{iterated degeneracies of elements of } \Delta_k^n \end{cases}$$

- The k^{th} horn $\wedge_k^n \subseteq \Delta^n$ is the subcomplex of Δ^n which is generated by the faces $d_j(\iota_n)$ except the k^{th} except the k^{th} face $d_k(\iota_n)$.

Coproduct Structure

Pullover,Pushout

4.7.2 Realization

Definition 4.7.2.1 Let $\mathcal{T}op$ denote the category of topological spaces. Then, define the realization functor

$$|| : S \rightarrow Top$$

A quick way to construct it which uses the simplex category $\Delta \downarrow X$ of the simplicial set X .

- Objects: maps

$$\sigma : \Delta^n \rightarrow X$$

- Morphisms: The following diagram commutes

$$\begin{array}{ccc} \Delta^n & & \\ \downarrow \theta & \searrow \sigma & \\ \Delta^m & \nearrow \tau & X \end{array}$$

is morphism.

◇ **Lemma 4.7.2.2** There exists an isomorphism

$$X \cong \lim_{\Delta^n \rightarrow X \in \text{Mor}(\Delta \downarrow X)} \Delta^n$$

- **Proof** The proof is the observation that any functor

$$C \rightarrow \mathbf{Sets}$$

, which is defined on a small category C , is a colimit of representable functors. $\square$

★ **Corollary 4.7.2.3** (Existence of Product in S) The product of simplicial sets exists. With this lemma, we have

Definition 4.7.2.4 Let X be a simplicial set. Then the **realization** $|X|$ of a simplicial set X is defined by the colimit

$$|X| = \lim_{\substack{\rightarrow \\ \Delta^n \rightarrow X \in \mathbf{Mor}(\Delta \downarrow X)}} |\Delta^n|$$

♠ **Proposition 4.7.2.5** The realization functor is left adjoint to the singular functor in the sense that there is an isomorphism

$$\mathbf{hom}_{Top}(|X|, Y) \cong \mathbf{hom}_S(X, S(Y))$$

which is natural in simplicial sets X and topological spaces Y .

- **Proof** We have

$$\begin{aligned} \mathbf{hom}_{Top}(|X|, Y) &\cong \lim_{\substack{\leftarrow \\ \Delta^n \rightarrow X}} \mathbf{hom}_{Top}(|\Delta^n|, Y) \\ &\cong \lim_{\substack{\leftarrow \\ \Delta^n \rightarrow X}} \mathbf{hom}_S(\Delta^n, S(Y)) \\ &\cong \mathbf{hom}_S(X, S(Y)) \end{aligned}$$

$\square$

Definition 4.7.2.6 Define: For simplicial set X ,

- $(sk_n X)$ is specified by:
 $(sk_n X)_m := \{x \in X_m \mid x = \alpha^*(y), y \in X_k, k \leq n, \alpha \in \mathbf{hom}_\Delta(m, k)\}$
- Skeleton Filtration

$$sk_0 X \subseteq sk_1 X \subseteq \dots$$

- Then, $X = \bigcup_{n \geq 0} sk_n X$.

Define the set of non-degenerate simplices of degree n

Definition 4.7.2.7

$$NX_n = \{x \in X_n \mid x \text{ not of the form } s_i y \text{ for any } 0 \leq i \leq n-1, y \in X_{n-1}\}$$

With pushout diagrams

$$\begin{array}{ccc} \sqcup_{x \in NX_n} \partial \Delta^n & \longrightarrow & sk_{n-1} X \\ \downarrow & & \downarrow \\ \sqcup_{x \in NX_n} \Delta^n & \longrightarrow & sk_n X \end{array}$$

Representation

Consider the realization of Δ^n is the space $|\Delta^n|$. Since $\Delta \downarrow \Delta^n$ has the terminal object

$$\sqcup_{0 \leq i < j \leq n} \Delta^{n-2} \xrightarrow{\rightarrow} \sqcup_{i=0}^n \Delta^{n-1} \rightarrow \partial \Delta^n$$

Given by the relation

...

Then, the coequalizer diagram of (topological) spaces

...

♠ **Proposition 4.7.2.8** $|X|$ is a CW-complex for each simplicial set X

◦ **Proof ...**

□

Definition 4.7.2.9 Take the notation: Category $CGHaus$ for compactly generated Hausdorff space.

♠ **Proposition 4.7.2.10** The functor $|| : S \rightarrow CGHaus$ preserves finite limits.

◦ **Proof ...**

□

4.7.3 Kan Complexes

Recall the representation of $\partial \Delta^n$

$$\sqcup_{0 \leq i < j \leq n} \Delta^{n-2} \xrightarrow{\rightarrow} \sqcup_{i=0}^n \Delta^{n-1} \rightarrow \partial \Delta^n$$

◇ **Lemma 4.7.3.1** The fork defined by the commutative diagram

$$\Delta^{n-2} \quad \Delta^{n-1}$$

◦ **Proof ...**

□

★ **Corollary 4.7.3.2** The set

$$\text{hom}_S(\wedge_k^n, X)$$

of simplicial set maps is in bijective correspondence with the set of n-tuples

...

◦ **Proof ...**

□

◇ **Lemma 4.7.3.3** For each space X , the map

$$S(X) \rightarrow * = \Delta^0$$

is a fibration.

◦ **Proof** ...

□

◇ **Lemma 4.7.3.4** (Moore) The underlying simplicial set of any simplicial group h IS FIBRANT.

◦ **Proof** ...

□

◇ **Lemma 4.7.3.5** ...

4.7.4 Anodyne Extensions

4.7.5 Function Complexes

Definition 4.7.5.1 Suppose X, Y be simplicial sets, the function complex $Hom(X, Y)$ is the simplicial set defined by

$$Hom(X, Y)_n := hom_S(X \times \Delta^n, Y)$$

If $\theta : m \rightarrow n$ is an ordinal number map, then the induced function

...

is defined by sending the map f to the composite

...

4.7.6 Simplicial Homotopy

Definition 4.7.6.1 Let $f, g : K \rightarrow X$ be simplicial maps. We say that there is a **simplicial homotopy**

$$f \xrightarrow{\simeq} g$$

from f to g if there is a commutative diagram

$$\begin{array}{ccc} K \times \Delta^0 \cong K & & \\ \downarrow 1 \times d^1 & \searrow f & \\ K \times \Delta^1 & \xrightarrow{h} & X \\ \uparrow 1 \times d^0 & \nearrow g & \\ K \times \Delta^0 = K & & \end{array}$$

Such h is called a **homotopy**.

Simplicial Homotopy Groups

4.7.7 Category of Fibrant Objects

4.7.8 Minimal Fibrations

4.7.9 The Closed Model Structure

More of Concepts in Model Categories will be given:

4.8 Lefschetz's Fixed Point Theorem

We have:

$$\operatorname{tr}(A) = \sum a_{ii} \quad \operatorname{tr}(AB) = \operatorname{tr}(BA)$$

4.8.1 Lefschetz Number

Definition 4.8.1.1

-
-

$$\Lambda(\varphi, F) := \sum_{n \in \mathbb{N}} (-1)^n \cdot \operatorname{tr}(\varphi_* : H_n(X, F) \rightarrow H_n(F))$$

Chapter 5 CW Complex

5.1 Cell Attachment

Definition 5.1.0.1 Let X, Y be topological space. Let $S \subseteq Y$ be a subspace and $\varphi : S \rightarrow X$ be a continuous function, then we write

$$X \cup_{\varphi} Y := X \cup_{X \leftarrow S} Y = (X \sqcup Y) / \sim$$

where $\sim$ is generated by $\varphi(s) \sim s, s \in S$.

Especially:

Definition 5.1.0.2 Let X be a topological space and $\varphi : S^{n-1} \rightarrow X$ be a continuous map. We consider the following continuous space

$$X \cup_{S^{n-1} \xrightarrow{\varphi} X} \bar{B}^n$$

with following language:

- We refer to the image of B^n as **open n -ball**;
- We refer to the image of $\bar{B}^n$ as **n -cell**;
- We refer to φ as the **attaching map** of the n -cell

We say: $X \cup_{\varphi} \bar{B}^n$ is obtained from X by attaching an n -cell along the attaching map φ .

♠ **Proposition 5.1.0.3** The map $X \rightarrow X \cup_{\varphi} \bar{B}^n$ is a closed embedding.

◦ **Proof** By topological push-out maps lemma. □

♣ **Example 5.1.0.4**

- Consider adding 0-cell: We have

$$\bar{B}_0 = \{0\}, S^{-1} = \emptyset$$

Thus,

$$X \cup_{S^{-1} \rightarrow X} \bar{B}^0 = X \sqcup \bullet$$

- Let $X = \bullet, \varphi : S^{n-1} \rightarrow \bullet$, then we have the following homeomorphism

$$\bullet \cup_{\bullet \leftarrow S^{n-1}} \bar{B}^{n-1} \xrightarrow{\cong} \bar{B}^n / S^{n-1} \xrightarrow{\cong} S^n$$

When $n = 2$, attaching the boundary to the point.

- Let $X = \bullet$ with attaching map $c : S^{n-1} \rightarrow \bullet$. If we attaching k n -cells to the space, we have

$$\underbrace{((\bullet \cup_c \bar{B}^n) \cup_c \dots) \cup_c \bar{B}^n}_k \cong S^n \vee \dots \vee S^n$$

Apply induction, we have

$$LHS \cong (S^n \vee \dots \vee S^n) \cup_c \bar{B}^n = (S^n \vee \dots \vee S^n) \vee S^n$$

- Consider the attaching of two balls via $\varphi : S^{n-1} \rightarrow \bar{B}^n$, we have

$$\bar{B}^n \cup_{\bar{B}^n \xleftarrow{\varphi} S^{n-1}} \bar{B}^n \xrightarrow{\cong} S^n$$

- Consider $X = S^1 \vee S^1$ with $\varphi : S^1 \rightarrow S^1 \vee S^1$.

5.1.1 Mapping Cone

Definition 5.1.1.1 Let X be a topological space with continuous $\varphi : S^{n-1} \rightarrow X$ continuous, then (The image is a cone, when $n = 2$)

$$\begin{aligned} \text{cone}(S^{n-1}) &:= (S^{n-1} \times [0, 1]) / (S^{n-1} \times \{0\}) \\ \text{cone}(f : S^{n-1} \rightarrow X) &:= (\text{Cone}(S^{n-1}) \sqcup X) / [(x, 1) \sim f(x)] \end{aligned}$$

◇ **Lemma 5.1.1.2** (Mapping Cone Attachment Lemma) The map

$$\begin{aligned} \text{Cone}(f : S^{n-1} \rightarrow X) &\rightarrow X \cup_{X \xleftarrow{f} S^{n-1}} \bar{B}^n \\ [P] &\mapsto \begin{cases} [P] & P \in X \\ [x \cdot r] & P = [(x, r)], (x, r) \in S^{n-1} \times [0, 1] \end{cases} \end{aligned}$$

is a homeomorphism.

- **Proof** Firstly, this is a homeomorphism

$$\begin{aligned} \text{Cone}(S^{n-1}) &\xrightarrow{\cong} \bar{B}^n \\ [(x, r)] &\mapsto x \cdot r \end{aligned}$$

Then, consider $\text{Cone}(S^{n-1}) \sqcup X \rightarrow \bar{B}^n \sqcup X$. Now the quotient by $f(x) \sim y, y \in S^{n-1}$, correspondent to $x \sim y$ in $\text{cone}(f)$. Thus homeomorphic. $\square$

5.1.2 Attaching Cell and Fundamental Group

♠ **Proposition 5.1.2.1** (Cell Attachment π_1 Proposition) Let X be a topological space, $n \geq 1$ with $\varphi : S^{n-1} \rightarrow X$ continuous. We set $x_0 := \varphi((0, \dots, 0, 1))$ the following statement holds:

- (n=1) We set $x_0 = \varphi(1), y_0 = \varphi(-1)$:
 - Suppose that x_0 and y_0 line in the same path-component of X , γ be a loop in $X \cup_\varphi \bar{B}^1$ that is given by the concatenation of obvious path in the 1-cell from x_0 to y_0 and some path in X from y_0 to x_0 , with the notation of the map

$$\pi_1(X, x_0) * \langle t \rangle \rightarrow \pi_1(X \cup_\varphi \bar{B}^1, x_0)$$

Given by the inclusion-induced map $\pi_1(X, x_0) \rightarrow \pi_1(X \cup_\varphi \bar{B}^1, x_0)$ and given by $[t] \mapsto [\gamma]$ is an isomorphism.

- Especially: Suppose x_0, y_0 lie in two different path components of X , then we can get an isomorphism

$$\pi_1(X, x_0) * \pi_1(X, y_0) \rightarrow \pi_1(X \cup_\varphi \bar{B}^1, x_0)$$

- (n>2) The inclusion-induced map $\pi_1(X, x_0) \rightarrow \pi_1(X \cup_\varphi \bar{B}^n, x_0)$ is an isomorphism;
- (n=2) The map $[0, 1] \rightarrow X$ given by

$$t \mapsto \varphi(\exp(2\pi it))$$

defines an element in $\pi_1(X, x_0)$, denote by $[\varphi]$. Then the elements in $\pi_1(X, x_0)$ denote by $[\varphi]$. Wwe have

$$\pi_1(X, x_0) / \langle [\varphi] \rangle \xrightarrow{\cong} \pi_1(X \cup_\varphi \bar{B}^2, x_0)$$

◦ **Proof**

- ($n = 1$) We consider the case $\varphi(1), \varphi(-1)$ in the same path-component of X . Then,
-
- ($n \geq 2$) Apply Seifert-van Kampen Theorem

...

□

Projective Space

◇ **Lemma 5.1.2.2** (Projective Space-Cell Attachment Lemma) Let $n \in \mathbb{N}$:

- We consider the continuous map

$$\begin{aligned} \varphi : S^{n-1} &\rightarrow \mathbb{R}P^{n-1} \\ (x_1, \dots, x_n) &\mapsto [x_1 : \dots : x_n] \end{aligned}$$

There exists a homeomorphism

$$\mathbb{R}P^{n-1} \cup_{\mathbb{R}P^{n-1} \xleftarrow{\varphi} S^{n-1}} \bar{B}^n \rightarrow \mathbb{R}P^n$$

such that the restriction to $\mathbb{R}P^{n-1}$ is given by the standard embedding

$$\mathbb{R}P^{n-1} \rightarrow \mathbb{R}P^n$$

- We think of $\bar{B}^{2n}$ and S^{2n-1} as subspace of $\mathbb{C}^n = \mathbb{R}^{2n}$. Consider the attaching map

$$\begin{aligned} \varphi : S^{2n-1} &\rightarrow \mathbb{C}P^{n-1} \\ (z_1, \dots, z_n) &\mapsto [z_1 : \dots : z_n] \end{aligned}$$

There exists a homeomorphism

$$\mathbb{C}P^{n-1} \cup_{\mathbb{C}P^{n-1} \xleftarrow{\varphi} S^{2n-1}} \bar{B}^{2n} \rightarrow \mathbb{C}P^n$$

such that the restriction to $\mathbb{C}P^{n-1}$ is given by the standard embedding

$$\mathbb{C}P^{n-1} \rightarrow \mathbb{C}P^n$$

◦ **Proof ...**

□

♠ **Proposition 5.1.2.3** (Projective Space π_1 -Proposition)

- We have $\pi_1(\mathbb{R}P^1) \cong \mathbb{Z}$, $\pi_1(\mathbb{R}P^n) \cong \mathbb{Z}_2$, $n \geq 2$;
- We have $\pi_1(\mathbb{C}P^n) = 0$, $n \in \mathbb{N}$.

◦ **Proof ...**

□

5.1.3 Retraction

◇ **Lemma 5.1.3.1** There exists an open neighborhood V of X in $X \cup_{\sqcup \varphi_i: S^{n_i-1} \rightarrow X} (\sqcup_i \bar{B}^{n_i})$ and a continuous map $r: V \rightarrow X$ that is the identity on X .

i.e. X is a neighborhood retract inside $X \cup_{\sqcup \varphi_i: S^{n_i-1} \rightarrow X} (\sqcup_i \bar{B}^{n_i})$.

◦ **Proof** Take $V := X \cup_{\sqcup \varphi_i} (\sqcup_i \bar{B}^{n_i} - \{0\})$, then define

$$\begin{aligned} r: V &\rightarrow X \\ x &\mapsto x, x \in X \\ (j, z) &\mapsto \varphi_j\left(\frac{z}{|z|}\right) \end{aligned}$$

□

$$\begin{array}{ccc} \sqcup_i S^{n_i-1} & \xrightarrow{\quad} & X \\ \downarrow & & \downarrow \\ \sqcup_i \bar{B}^{n_i} & \xrightarrow{\quad} & X \cup_{\sqcup \varphi_i} (\sqcup_i \bar{B}^{n_i}) \supseteq V \end{array} \quad \begin{array}{c} \uparrow \\ r \end{array}$$

5.1.4 Attaching Arbitrary Many Cells

Definition 5.1.4.1 Let X be a topological space and let $\{\varphi_i: S^{n_i-1} \rightarrow X\}_{i \in I}$ be a family of continuous maps:

- We consider the topological space

$$X \cup_{\sqcup \varphi_i: X \leftarrow S^{n_i-1}} (\sqcup_i \bar{B}^{n_i})$$

- $S^{n-1} \rightarrow \bar{B}^n$ are closed embeddings together with the topological pushout lemma, the natural map of X into above topological space

$$X \rightarrow X \cup_{\sqcup \varphi_i: X \leftarrow S^{n_i-1}} (\sqcup_i \bar{B}^{n_i})$$

is a closed embedding.

◇ **Lemma 5.1.4.2**

- The composite $X \xrightarrow{\text{incl}} X \sqcup (\sqcup_i \bar{B}^{n_i}) \xrightarrow{p} X \cup_{\sqcup \varphi_i} (\sqcup_i \bar{B}^{n_i})$ is a homeomorphism onto its image $p(X)$ which is closed in $X \cup_{\sqcup \varphi_i} (\sqcup_i \bar{B}^{n_i})$;
- The composite $\sqcup B^{n_i} \xrightarrow{\text{incl}} X \sqcup (\sqcup_i B^{n_i}) \xrightarrow{p} X \cup_{\sqcup \varphi_i} (\sqcup_i B^{n_i})$ is a homeomorphism onto its image $p(\sqcup_i S^{n_i-1})$, which is an open subset of $X \cup_{\sqcup \varphi_i} (\sqcup_i \bar{B}^{n_i})$;
- As a set, $X \cup_{\sqcup \varphi_i} (\sqcup_i \bar{B}^{n_i})$ is the disjoint union of $p(X)$ and $p(\sqcup_i B^{n_i})$

◦ **Proof**

-
-
-

□

◇ **Lemma 5.1.4.3** (Iterative Cell-Attachment Lemma) Let X be a topological space and let $\{\varphi_i : S^{n_i} \rightarrow X\}_{i \in I}$ be a family of continuous maps, $J \subseteq I$ be a subset. We denote by X_J the result attaching the cell in J and we by X_I the result of attaching the cells in I :

- $X_J \rightarrow X_I$ is a closed embedding;
- If we attach the cells corresponding to $J - I$ to X_J , then the resulting topological space is naturally homeomorphic to X_I .

◦ **Proof**

- By
-

□

5.1.5 Cell Attachment-Hausdorff Proposition

♠ **Proposition 5.1.5.1** (Cell Attachment Hausdorff Proposition) Let X be a topological space and let $\{\varphi_i : S^{n_i-1} \rightarrow X\}$ be a family of continuous maps. We consider the topological space

$$\hat{X} := X \cup_{\sqcup_{i \in I} \varphi_i : X \leftarrow \sqcup_{i \in I} S^{n_i-1}} \sqcup_{i \in I} \bar{B}^{n_i}$$

That we obtain from X by attaching cells along the attaching maps φ_i :

- Suppose $U, V \subseteq X$ be disjoint open subsets of X , then there exist disjoint open subsets $\hat{U}$ and $\hat{V}$ of $\hat{X}$ with

$$X \cap \hat{U} = U \quad X \cap \hat{V} = V$$

- If X is Hausdorff, so is $\hat{X}$.

◦ **Proof** Define:

$$\Phi_i : \bar{B}^{n_i} \rightarrow \hat{X}$$

Firstly, $\hat{U} \subseteq \hat{X}$ is open $\Leftrightarrow$ The intersection $\hat{U} \cap X$ is open in X , for all $i \in I$ the preimage $\Phi_i^{-1}(\hat{U})$ is open.

We could construct $\hat{U}, \hat{V}$ via

$$\hat{U} := U \cup (\cup_{i \in I} \Phi_i(\{r \cdot u | u \in \Phi_i^{-1}(U), r \in (1/2, 1]\}))$$

Such $\hat{U}$ is open:

- $X \cap \hat{U}$: Consider each

$$\Phi_i(\{r \cdot u | u \in \Phi_i^{-1}(U), r \in (1/2, 1]\}) \cap X = \Phi_i(\Phi_i^{-1}(U)) \cap X = \{\varphi_i(u) | u \in \Phi_i^{-1}(U)\}$$

And we have: $\Phi_i^{-1}(U) \cap S^{n_i-1} = \{u \in S^{n_i-1} | \varphi_i(u) \in U\} \subseteq \text{Im}(\varphi) \cap U$.

We have: $\hat{U} \cap X = U$ is open;

- Consider $\Phi_j^{-1}(\hat{U})$:

– X : Boundaries $u \in \bar{S}^{n_i-1}, \varphi_j(u) \in U$;

- Cell $\bar{B}_j$: Points $r \cdot u, r \in (1/2, 1), \varphi_j(u) \in U$
- Other Cells: No.

Thus

$$\Phi_j^{-1}(\hat{U}) \xrightarrow{\cong} (1/2, 1] \times \varphi_j^{-1}(U)$$

which is open in $\bar{B}^{n_j}$.

And $\hat{U} \cap \hat{V} = \emptyset$:

- $\hat{U} \cap \hat{V} \cap X = U \cap V = \emptyset$;
- In cell: If $x \in \bar{B}^{n_i} \cap \hat{U} \cap \hat{V}$, we have

$$x = \varphi_i(r \cdot u), \varphi_i(u) \in U \cap V$$

contradict to $U \cap V = \emptyset$.

This is Hausdorff: Select U, V such such that $U \cap V = \emptyset$. □

Another proof by Note of Topology 1, via retraction map

◦ **Proof** Suppose $x \neq y \in \hat{X}$.

- $x, y \in$;
- $x \in X, y \in$;
- $x, y \in X$. X is Hausdorff,

□

Furthermore, we have

♠ **Proposition 5.1.5.2**

- If X is compact, I is finite, then $\hat{X} := X \cup \sqcup_{i \in I} \varphi_i : X \leftarrow \sqcup_{i \in I} S^{n_i-1} \sqcup_{i \in I} \bar{B}^{n_i}$ is finite;
- Let K be quasi-compact subset of $\hat{X} := X \cup \sqcup_{i \in I} \varphi_i : X \leftarrow \sqcup_{i \in I} S^{n_i-1} \sqcup_{i \in I} \bar{B}^{n_i}$, then

$$K \cap B^{n_i} = \emptyset$$

for almost all $i \in I$. In particular, if I is infinite, then $\hat{X} := X \cup \sqcup_{i \in I} \varphi_i : X \leftarrow \sqcup_{i \in I} S^{n_i-1} \sqcup_{i \in I} \bar{B}^{n_i}$ is not compact.

◦ **Proof**

-
-

□

5.1.6 Cell Attachment-Normal Proposition

◇ **Lemma 5.1.6.1** (Auxiliary Lemma) Let $U, V \subseteq S^{n-1}$ be disjoint open subsets and $B \subseteq \bar{B}^n$ be a closed subset with $B \cap S^{n-1} \subseteq V$. There exists an $\epsilon > 0$ such that

$$\{r \cdot u | r \in (1 - \epsilon, 1], u \in U\} \cap B = \emptyset$$

◦ **Proof ...** □

♠ **Proposition 5.1.6.2** (Cell-Attachment Normal Proposition) Let X be a topological space and let $\{\varphi_i : S^{n_i-1} \rightarrow X\}_{i \in I}$ be a family of continuous maps. We consider the topological space

$$\hat{X} := X \cup_{\sqcup_{\varphi_i : X \leftarrow \sqcup_{i \in I} S^{n_i-1}} \sqcup_{i \in I} \bar{B}^{n_i}}$$

We obtain from X by attaching cells along the attaching maps φ_i :

- Let $\hat{A}, \hat{B} \subseteq \hat{X}$ be two disjoint closed subsets. We set

$$A := X \cap \hat{A}, B := X \cap \hat{B}$$

if there exist disjoint open neighborhoods $U \subseteq X, V \subseteq Y$ of A, B then there exist disjoint open neighborhoods $U \subseteq X, V \subseteq Y$ of A and B then there exists disjoint open neighborhood $\hat{U} \subseteq \hat{X}, \hat{V} \subseteq \hat{X}$ of $\hat{A}, \hat{B}$ with

...

- If X is normal, then $\hat{X}$ is also normal.

◦ **Proof ...** □

5.2 CW Complex

Relative CW Complex

Definition 5.2.0.1 A **relative CW complex** for a pair (X, A) is a topological space together with filtration

$$A = X_{-1} \subseteq \cdots \subseteq X$$

with:

- (Cell Structure) For every $n \in \mathbb{N}$, there exists a pushout of topological space

$$\begin{array}{ccc} \sqcup_{i \in I_n} S^{n-1} & \xrightarrow{\sqcup} & X_{n-1} \\ \downarrow \sqcup j_n & & \downarrow \\ \sqcup_{i \in I_n} \bar{B}^n & \longrightarrow & X_n \end{array}$$

- (Direct-Limit Topology)

5.2.1 CW Complex

Definition 5.2.1.1 A CW-complex X is a topological space together with a filtration

$$\emptyset = X^{-1} \subseteq X^0 \subseteq \dots \subseteq X$$

by subspaces which has the following properties:

- We have $X = \text{colim}_{\rightarrow} X^n$, as described in the topological-direct colimit proposition. i.e. As a set we have

$$X = \bigcup_{n \in \mathbb{N}_0} X^n$$

And $U \subseteq X$ is open $\Leftrightarrow$ for each $n \in \mathbb{N}_0$, the intersection $U \cap X^n$ is open in X^n .

- For each $n \in \mathbb{N}_0$, there exists a set I , a family $\{\varphi_i : S^{n-1} \rightarrow X^{n-1}\}_{i \in I}$ of continuous maps and a homeomorphism

$$\Xi : X^{n-1} \cup_{\sqcup \varphi_i : X^{n-1} \leftarrow \sqcup_{i \in I} S^{n-1}} \sqcup_{i \in I} \bar{B}^n \rightarrow X^n$$

which is the inclusion on X^{n-1} .

Then:

Definition 5.2.1.2 Using the notation from above:

- X^n is called **n-skeleton** of X ;
- The component $X^n - X^{n-1}$ is called the **open n-cells** of X ;
- For each n -cell σ of X we refer to its closure $\bar{\sigma}$ in X as an **n-cell** of X
- $\dim(X) := \sup\{n \in \mathbb{Z}_{\geq -1} \mid X^{n-1} \neq X^n\}$
- The CW-complex is **finite** if it has only finite many cells;
and we say that it is countable if it has countably many cells;

♠ **Proposition 5.2.1.3** Consider the set-theoretic image of filtration $|X|$, naturally has a family of maps $\Phi_i : \sqcup_{J_i} \bar{B}_i \rightarrow |X|$. Then: The following topologies are equivalent on $|X|$:

- The initial topology induced by $\{\Phi_i\}_{i \in \mathbb{N}}$;
- The weak topology defined above.

◦ **Proof** Notice that: The two topologies on $|X|$.

- $\geq$: Suppose U is open in weak topology $\Leftrightarrow U \cap X^n$ is open, $n \in \mathbb{N} \Rightarrow \Phi_n^{-1}(U)$
- $\leq$:

□

Cells

Definition 5.2.1.4 Let X be a CW-complex:

- Given $n \in \mathbb{N}$, we denote by $\text{Cell}_n(X) :=$. Furthermore

$$\text{Cell}(X) = \bigcup_{n \in \mathbb{N}} \text{Cell}_n(X)$$

- Let $c \in \text{Cell}_n(x)$ be an n -cell of X :

- An **attaching map** for c is a continuous map

$$S^{n-1} \rightarrow X^{n-1}$$

such that there exists a homeomorphism

$$\Phi_c : X^{n-1} \cup_{\varphi: S^{n-1} \rightarrow X^{n-1}} \bar{B}^n \rightarrow X^{n-1} \cup c \subseteq X^n$$

which is the identity on X^{n-1} . (i.e. $\Phi_c|_{X^{n-1}} = id_{X^{n-1}}$)

- A **characteristic map** for c is the continuous map $\Phi : \bar{B}^n \rightarrow X$ such that

*

$$X^{n-1} \cup_{\Phi|_{S^{n-1}}: S^{n-1} \rightarrow X^{n-1}} \bar{B}^n \rightarrow X^{n-1} \cup c$$

induced an identity on X^{n-1} ;

- * The map Φ on $\bar{B}^n$ is well-defined, and it is a homeomorphism (onto its image).

- A **full set of attaching maps for the cell of CW-complex X** is defined as a family of continuous maps

$$\{\varphi_c : S^{dim(c)-1} \rightarrow X^{dim(c)-1}\}_{c \in Cell(X)}$$

such that: each φ_c is an attaching map for the cell c .

Similarly, define the fullset of characteristics maps for the n -cells of X .

Definition 5.2.1.5 (P2136,Reg) Let: X be a CW-complex. Then we know that for $n \in \mathbb{N}$, there exists a family

$$\{\varphi_c : S^{n-1} \rightarrow X^{n-1}\}_{c \in Cell_n(X)}$$

of continuous maps and a homeomorphism

$$\Xi_n : X^{n-1} \cup_{\sqcup \varphi_c: \sqcup_c S_c^{n-1} \rightarrow X} (\sqcup_{c \in Cell(X)} \bar{B}_c^n) \rightarrow X^n$$

which is the inclusion on X^{n-1} . Then for $c \in Cell_n(X)$ consider the map

$$\begin{aligned} \Phi_c : \bar{B}^n &\rightarrow X^{n-1} \cup c \\ x &\mapsto \Xi_n([(x, c)]) \end{aligned}$$

Cell-Compactness

♠ **Proposition 5.2.1.6** Let X be a CW-complex:

- Let c be an n -cell of X . If $\Phi : \bar{B}^n \rightarrow X$ is a characteristic map for vc , then

$$\Phi(\bar{B}^n) = c$$

- Every cell is compact.

◦ Proof

-
-

□

♠ **Proposition 5.2.1.7** (Characteristic Maps Proposition) Let X be a CW-complex. Furthermore let

$$\{\Phi_c : \bar{B}^{dim(c)} \rightarrow X^{dim(c)}\}_{c \in Cell(X)}$$

be a full set of characteristic maps for the cells of X

-
-

◦ **Proof ...**

□

Characteristic Maps

♠ **Proposition 5.2.1.8** (Characteristic Maps Proposition) Let X be a CW-complex, $\{\Phi_c : \bar{B}^{dim(c)} \rightarrow X^{dim(c)}\}_{c \in Cell(X)}$ be a full set of characteristic maps for the cells of X :

- Given any subset $U \subseteq X$ the following statements are equivalent:
 -
 -
 -
- Let $f : X \rightarrow Y$ be a map to a topological space Y . The following statements are equivalent:
 - f is continuous;
 - For every $c \in Cell(X)$ the restriction of f to c is continuous;
 - For every $c \in Cell(X)$, the map $f \circ \Phi_c : \bar{B}^{dim(c)} \rightarrow Y$ is continuous;

◦ **Proof ...**

□

Subcomplex

Definition 5.2.1.9 Suppose X be a CW-complex. Then, a subcomplex $A \subseteq X$ is a CW-complex A which satisfies

- For each $n \in \mathbb{N}$ we have

$$A^n = A \cap X^n$$

- Each open n -cell of A is also an open n -cell of X ;

◇ **Lemma 5.2.1.10** (Subcomplex-Closed Lemma) Each subcomplex of a CW-complex X is a closed subset of X

◦ **Proof Claim 1** Let a be an open n -cell of A . The closure of a in A equals the closure of a in X .

Claim 2 If A intersects an open n -cell x of X , then $\bar{x}$ is an n -cell of A .

Claim 3 For each $n \in \mathbb{N}$ the intersection $A \cap X^n$ is a closed subset of X^n .

□

♠ **Proposition 5.2.1.11** (Subcomplex-Characterization Proposition)

◦ **Proof ...**

□

♠ **Proposition 5.2.1.12** [Exercise 109.9, Reg](CW-Subcomplex Criteria for Connected CW-Complex) Let $A \subseteq X$ be a subspace. We suppose that A has the following properties: For any open cell c of X with $c \cap A \neq \emptyset$, we have

$$\bar{c} \subseteq A$$

◦ **Proof ...**

□

♠ **Proposition 5.2.1.13** (Component-Subcomplex Lemma) Let X be a CW-complex. Every (path-) component of X is a subcomplex of X .

◦ **Proof ...**

□

♠ **Proposition 5.2.1.14** (CW-Complex Pasting Lemma) Let X be a CW-complex and let $\{X_i\}$ be a family of subcomplexes such that

$$X = \cup X_i$$

The Finiteness Theorem for CW-Complexes

► **Theorem** (CW-Complex Finiteness Theorem) Every compact subset of a CW-complex X is contained in a finite subcomplex.

◦ **Proof ...**

□

★ **Corollary 5.2.1.15** A CW-complex is compact iff it is finite

◦ **Proof ...**

□

◇ **Lemma 5.2.1.16** ...

◦ **Proof ...**

□

Topological Graphs

◇ **Lemma 5.2.1.17** (Graph-CW Complex Lemma)

- The graph $(X, G = (V, E, \theta))$ is an undirected topological graph. Then, $\emptyset \subseteq$
-

Smooth Manifolds

► **Theorem** (Smooth Manifold CW-Theorem) Every text

◦ **Proof ...**

□

Fundamental Groups

◇ **Lemma 5.2.1.18** (CW-Skeleton- π_0 -Lemma) Let X be a CW-complex.

- The inclusion-induced map $\pi_0(X^0) \rightarrow \pi_0(X)$ is a surjection;
- For every $k \geq 1$ the inclusion-induced map $\pi_0(X^k) \rightarrow \pi_0(X)$ is a bijection.

◦ **Proof ...**

□

Haudorff Property

Normal Proposition

Second-Countability

Metrizability

Partition of Unity

Paracompactness

CW-Structure of Δ^k

♠ **Proposition 5.2.1.19** Let $k \in \mathbb{N}$: Consider the filtration of the standard k -simplex Δ^k that is given by

$$(\Delta_k)^i := \{(t_0, \dots, t_k) \in \Delta^k \mid \text{at least } k-i \text{ coordinates are zero}\}, i = 0, \dots, k$$

Then,

◦ **Proof ...**

□

Numbers of CW-Structure

Definition 5.2.1.20 Given a compact topological space X that admits a CW-structure. We define:

...

♠ **Proposition 5.2.1.21** We have:

- $\Theta(S^n) =$;
- $\Theta(S^1 \times S^1) =$;
- $\Theta(\mathbb{R}P^2) =$;
- $\Theta(\text{Klein bottle})$;
- Let $n \in \mathbb{N}$, then $\Theta(\bar{B}^n) =$

◦ **Proof ...**

□

5.2.2 Homotopies

Topology of the Product of a CW-Complex with Interval

♠ **Proposition 5.2.2.1** (CW-Interval Proposition) Let X be a CW-complex

- The following three topologies on $X \times [0, 1]$ agree:
 - The product topology on the product $X \times [0, 1]$;
 - The topology on $|X \times [0, 1]$ viewed as the direct colimit

$$\xrightarrow{\text{colim}} (X^k \times [0, 1])$$

- The topology which is defined on the set $X \times [0, 1]$ by the condition:(i.e. The weak topology induced by $\Phi \times id_{[0,1]}$)
- Let $f : X \times [0, 1] \rightarrow Y$ be a map to a topological space Y . The followings are equivalent:
 - The map f is continuous if $X \times [0, 1]$ is equipped with the product topology;
 - For every $k \in \mathbb{N}$, the restriction of f to $X^k \times [0, 1]$ is continuous;
 - For every characteristic map $\Phi : \bar{B}^n \rightarrow X$ of cell of X the map

$$\bar{B}^n \times [0, 1] \xrightarrow{(v,t) \mapsto (\Phi(v),t)} X \times [0, 1] \xrightarrow{f} Y$$

is continuous.

◦ **Proof ...**

□

◇ **Lemma 5.2.2.2** (Infinite Sphere-Contractibility Lemma) The infinite-dimensional sphere S^∞ is contractible.

◦ **Proof ...**

□

Trees

♠ **Proposition 5.2.2.3** (Iterated Deformation Retract Proposition) Let X be a CW-complex, and $Y_0 \subseteq Y_1 \subseteq \dots$ be a sequence of subcomplexes of X such that the two conditions satisfied:

-
-

◦ **Proof ...**

□

Definition 5.2.2.4 ...

The fundamental group of 1-dimensional CW-complexes

Local Properties of CW-Complexes

♠ **Proposition 5.2.2.5** ...

Seifert-van-Kampen Theorem

5.2.3 Coverings of CW-Complexes

Coverings of CW-Complex

Subgroups of Finitely Presented Groups

Subgroups of Free Groups

5.2.4 Constructions

Two Categories

Definition 5.2.4.1 Define:

Cellular

Definition 5.2.4.2 ...

Functor from CW-complexes to topological spaces

◇ **Lemma 5.2.4.3** ...

CW-Disjoint Union

♠ **Proposition 5.2.4.4** Let $\{X_i\}$ be a family of CW-complexes

-
-
-

Glueing

Quotient

Direct Colimit

Pushout

Definition 5.2.4.5 ...

Products

Cones, Suspensions

Definition 5.2.4.6 Let X be a CW-complex:

- The cone

$$\text{cone}(X) := (X \times [0, 1]) / (X \times \{0\})$$

- The suspension

$$\Sigma(X) := (X \times [-1, 1]) / (X \times \{-1\}) / (X \times \{1\})$$

♠ **Proposition 5.2.4.7** (CW-Complex-Cone and Suspension Proposition)

- The cone $\text{Cone}(X)$ admits a CW-structure, such that the inclusion map $X \rightarrow \text{Cone}(X)$ given by

- The suspension $\Sigma(X)$ admits a CW-structure, such that the inclusion map $X \rightarrow \Sigma(X)$ given by

◦ **Proof**

□

Definition 5.2.4.8 ...

♠ **Proposition 5.2.4.9** ...

◦ **Proof** ...

□

CW-HNN-van-Seifert Theorem

5.2.5 Homotopy Extension Theorem

The Homotopy Extension Theorem

Definition 5.2.5.1

- A continuous map $i : A \rightarrow X$ between topological spaces is called a **cofibration** if it has the **homotopy extension property**. i.e. if given any topological space Y and two continuous maps

$$f : X \times \{0\} \rightarrow Y \quad G : A \times [0, 1] \rightarrow Y$$

with $f(i(a), 0) = G(a, 0)$, $a \in A$. There exists a continuous map

$$F : X \times [0, 1] \rightarrow Y$$

such that the following properties diagram commutes

$$\begin{array}{ccccc}
 A \times \{0\} & \hookrightarrow & & A \times [0, 1] & \\
 \downarrow i \times id_{\{0\}} & & \nearrow G & \downarrow i \times id_{[0, 1]} & \\
 & & Y & & \\
 \downarrow & \nearrow f & & \nwarrow F & \\
 X \times \{0\} & \hookrightarrow & & X \times [0, 1] &
 \end{array}$$

- If $i : A \rightarrow X$ is a cofibration such that $i(A)$ is a closed subset of X , then we refer to i as a **closed fibration**.

► **Theorem** (Homotopy Extension Theorem) Let X be a CW-complex and let A be a subcomplex of X :

- The inclusion map $A \rightarrow X$ is a closed cofibration;
- In slightly different word: Let Y be a topological space and let $f_0 : X \rightarrow Y$ be a continuous map. Furthermore, suppose that $A \subseteq X$ is a subcomplex, and suppose we are given a homotopy $G : A \times [0, 1]$ from $g_0 := f_0|_A$ to some map $g_1 : A \rightarrow Y$, there exists a homotopy

$$F : X \times [0, 1] \rightarrow Y$$

starting from f_0 such that the restriction $F|_{A \times [0, 1]} = G$.

◦ **Proof ...**

□

★ **Corollary 5.2.5.2** (CW-Complex-Vertices-Homotopy Corollary) Let X be a connected CW-complex, let P be a point in the 0-skeleton of X and let Q be some other point on Y :

- Given any path $\gamma : [0, 1] \rightarrow X$ with

$$\gamma(0) = P \quad \gamma(1) = Q$$

There exists a homotopy $F : X \times [0, 1] \rightarrow X$ with

$$F(P, t) = \gamma(t)$$

- There exists a continuous map

$$h : X \rightarrow X$$

with $h(P) = Q$ that is homotopic to id_X .

◦ **Proof ...**

□

★ **Corollary 5.2.5.3** (Quotient by Contractible Corollary) Let X be a CW-complex, $A \subseteq X$ be a subcomplex

-
-

5.2.6 Cellular Approximation Theorem Theorem

Definition 5.2.6.1 A continuous map $g : X \rightarrow Y$ between CW-complexes is called **cellular** if for each $n \in \mathbb{N}$, we have

$$f(X^n) \subseteq Y^n$$

► **Theorem** (CAT) Let (X, A) and (Y, B) be relative CW-complexes and $f : X \rightarrow Y$ be continuous with $f(A) \subseteq B$, then f is homotopic rel A to a cellular map.

◦ **Proof ...**

□

CW-Skeleton- π_n -Proposition

♠ **Proposition 5.2.6.2** Let X be connected CW-complex, and x_0 be a point in 0-skeleton of X and $j \in \mathbb{N}$:

-
-

◦ **Proof ...**

□

♠ **Proposition 5.2.6.3** (Sphere Low π_n Proposition) Let $n \in \mathbb{N}$. For any $k \in \{1, \dots, n-1\}$ we have

...

◦ **Proof ...** □

Cellular Approximation Theorem for Pairs

► **Theorem** Suppose we are given a continuous map

$$g : (X, A) \rightarrow (Y, B)$$

between pairs of CW-complexes. Let Z be a subcomplex of A that the restriction of g to Z is a cellular map, then g is homotopic rel Z to a cellular map

$$\hat{g} : (X, A) \rightarrow (Y, B)$$

◦ **Proof ...** □

5.3 The CW-Approximation Theorem

5.3.1 k -Connected

Firstly:

- Given $n \in \mathbb{N}$ we write

$$* = (0, \dots, 0, 1) \in S^n$$

Then, we can view S^n as a CW-complex with one 0-cell $\{*\}$ and one n -cell.

- We could form any wedge of spheres

$$\bigvee_{i \in I} S_i^{n_i}$$

by identifying the points $*$. Then, it can be viewed as a CW-complex by the CW-Wedge lemma.

We say:

Definition 5.3.1.1 Let $f : X \rightarrow Y$ be a continuous map between topological spaces:

- We say that the map $f : X \rightarrow Y$ is 0-connected if $f_* : \pi_0(X) \rightarrow \pi_0(Y)$ is a surjection.
In other words,
- Let $k \in \mathbb{N}$. we say $f : X \rightarrow Y$ is **k -connected** if the following two conditions are satisfied:
 - The map $f_* : \pi_0(X) \rightarrow \pi_0(Y)$ is a bijection;
 - For every $x_0 \in X$, the induced map

$$f_* : \pi_i(X, x_0) \rightarrow \pi_i(Y, f(x_0)) \text{ is an } \begin{cases} \text{isomorphism} & i = 1, \dots, k-1 \\ \text{epimorphism} & i = k \end{cases}$$

♠ **Proposition 5.3.1.2** (Higher Connected Map Proposition) Let $k \in \mathbb{N}$, A be a CW-complex and let $f : A \rightarrow X$ be continuous map.

If $A \rightarrow X$ is k -connected, there exists a CW-complex B and a continuous map $g : B \rightarrow X$ such that the following statement hold:

- A is a subcomplex of B ;
- The only extra cells of B are $(k + 1)$ -dimensional.
In particular the k -skeleton of B equals the k -skeleton of A ;
- the restriction of g to A equals f ;
- The map $g : B \rightarrow X$ is $(k+1)$ -connected

◦ **Proof** Consider the following diagram

$$\begin{array}{ccc} k\text{-dim CW-complex } A & \xrightarrow{k\text{-connected } f} & X \in \mathcal{Top} \\ \downarrow & \nearrow (k+1)\text{-connected } g & \\ (k+1)\text{-dim CW-complex } B, B^k = A & & \end{array}$$

Problem for f : Write $x_0 = f(a_0)$

- The map $\pi_k(A, a_0) \rightarrow \pi_k(X, x_0)$ is not a monomorphism;
- The map $\pi_{k+1}(A, A_0) \rightarrow \pi_{k+1}(X, x_0)$ is not an epimorphism.

We construct the map $g : B \rightarrow X$ as follow:

- By the CW-Skeleton proposition,

$$\pi_k(A^k, x_0) \rightarrow \pi_k(A, a_0)$$

is an epimorphism. This implies that we can pick a family

$$\{\varphi_i : (S^k, *) \rightarrow (A^k, a_0)\}$$

-
-

□

5.3.2 Weak Homotopy Equivalence and the CW-Approxiamation Theorem

Weak Homotopy Equivalence

Definition 5.3.2.1 We say map $f : X \rightarrow Y$ between topological spaces is a **weak homotopy equivalence** if f is k -connected for all $k \in \mathbb{N}$.

◇ **Lemma 5.3.2.2** (Weak Homotopy Equivalence Criteria Lemma) Let $f : X \rightarrow Y$ be a continuous map between topological spaces:

- If f is a homotopy equivalence, then it is also a weak homotopy equivalence:
- If $g : X \rightarrow Y$ is a continuous map that homotopic to f , then

f is a weak homotopy equivalence $\Leftrightarrow g$ is a weak homotopy equivalence

- If X, Y are 0-connected, then following holds:

f is weak homotopy equivalence $\Leftrightarrow$ There exists an $x_0 \in X$ such that for every $k \in \mathbb{N}$, the induced map $f_* : \pi_k(X, x_0) \rightarrow \pi_k(Y, f(x_0))$ is an isomorphism

◦ **Proof ...** □

♠ **Proposition 5.3.2.3** ...

CW-Approximation

Definition 5.3.2.4 Let X be a topological space. A **CW-Approximation** is a CW-Complex Z with a weak homotopy equivalence $f : Z \rightarrow X$.

◇ **Lemma 5.3.2.5** (Product-CW Approximation) Let X, Y be two CW-complexes. Let P be the product topology on $X \times Y$, and $CWWP$ be the CW-product topology. Then, identity map

$$id : (X \times CWP) \rightarrow (X \times Y, P)$$

is a CW-approximation.

◦ **Proof ...** □

► **Theorem** (CW-Approximation Theorem)

- Every topological space X admits a CW-approximation $f : Z \rightarrow X$;
- Let $k \in \mathbb{N}$. If X is k -connected space, then there exists a CW-approximation

$$f : Z \rightarrow X$$

such that Z has single 0-cell and such that Z has no cells in dimension $1, \dots, k$.

In both case, given a point $x_0 \in X$, we can find f and $z_0 \in Z^0$ such that

$$f(z_0) = x_0$$

5.4 The Whitehead Theorem

◇ **Lemma 5.4.0.1** (Compression Lemma) Let (X, A) be a pair of CW-complexes and let (Y, B) be a pair of topological spaces with $B \neq \emptyset$.

◦ **Proof ...** □

► **Theorem**

(Whitehead Theorem) Let $f : X \rightarrow Y$ be a continuous map between two connected CW-complexes:

- If $f : X \rightarrow Y$ is a weak equivalence, then f is a homotopy equivalence.

In fact, there is more refined version:

- Let $x_0 \in X^0, y_0 \in Y^0$ be the base points in 0-skeleta of X respectively Y . If $f : X \rightarrow Y$ is a equivalence with $f(x_0) = y_0$, then there exists a homotopy equivalence between the pointed topological spaces (X, x_0) and (Y, y_0) .

◦ **Proof ...**

□

5.4.1 Mapping Cylinder

Definition 5.4.1.1 Given a continuous map $f : X \rightarrow Y$ between topological spaces, define

$$Cyl(f) := Cyl(f : X \rightarrow Y) := ((X \times [0, 1]) \sqcup Y) / \sim \quad (x, 1) \sim f(x), x \in X$$

5.4.2 Homotopy Classification of Lens Spaces

Definition 5.4.2.1 Consider the action of $\mathbb{Z}_p$ on $S^3 \subseteq \mathbb{C}^2$: Suppose $\gcd(p, q) = 1$, then the mmap

5.5 Eilenberg-MacLane Space

5.5.1 Moore Space

Recall the definition of Moore space

Definition 5.5.1.1 Let π be an abelian group and $n \in \mathbb{N}$, we say that a non-empty topological space X is a **Moore space** of type $M(\pi, n)$ if the following conditions are satisfied:

- X admits a CW-structure;
- For any $i \neq n$, we have

$$\hat{H}_i(X, \mathbb{Z}) = 0$$

- $H_n(X) \cong \pi$;
- If $n \geq 2$, then we demand that X is simply connected;

♠ **Proposition 5.5.1.2** (Moore Space Existence Proposition) Given any $n \in \mathbb{N}$ and given any abelian group π , there exists an $(n + 1)$ -dimensional CW-complex that has no cells in dimension.

◦ **Proof ...**

□

5.5.2 Eilenberg-MacLane Space

Definition 5.5.2.1 Let π be a group and $n \in \mathbb{N}$. Let X be a topological space:

- We say that X is an **Eilenberg-MacLane space** of type $K(\pi, n)$ if the following conditions hold
 - X admits CW-structure;
 - For any $i \in \mathbb{N}$ with $i \neq n$ we have $\pi_i(X) = 0$;
 - $\pi_n(X) \cong \pi$;

- Let $x_0 \in X$. We say

Chapter 6 Homotopy Theory

6.1 Homotopies

Definition 6.1.0.1 Let X, Y be two continuous maps:

- Let $f_0, f_1 : X \rightarrow Y$ be continuous maps. Then, a **homotopy** between f_0 and f_1 is a continuous map

$$F : X \times [0, 1] \rightarrow Y$$

with $F_0 = f_0, F_1 = f_1$. When there exists a homotopy between the maps f_0 and f_1 , then we say that f_0 and f_1 are **homotopic**, write

$$f_0 \simeq f_1$$

- $f : X \rightarrow Y$ is **null-homotopic** iff $f \simeq c$ where $c : X \rightarrow Y$ is a constant map.

Homotopies in Convex Subset of $\mathbb{R}^n$

◇ **Lemma 6.1.0.2** Let X be a topological space, given any two continuous maps $f_0, f_1 : X \rightarrow Y$ to a convex subset $Y \subseteq \mathbb{R}^n$, the map

$$\begin{aligned} F : X \times [0, 1] &\rightarrow Y \\ (x, t) &\mapsto f_0(x) \cdot (1 - t) + f_1(x) \cdot t \end{aligned}$$

is a homotopy between f_0 and f_1 .

◦ **Proof ...**

□

Homotopy Rel Subset

Definition 6.1.0.3

-
-

Isotopy, Homeotopy

Definition 6.1.0.4 ...

◇ **Lemma 6.1.0.5** (Convex-Isotopic-to-Ball Lemma)

◦ **Proof ...**

□

◇ **Lemma 6.1.0.6** (Homotopy Combination Lemma)

◦ **Proof ...**

□

Homotopy Quotient, Pushout

► Theorem ...

◦ Proof ...

□

6.1.1 Homotopy Equivalences and Homotopy Equivalent Topological Spaces

Deformation Retractions

Definition 6.1.1.1 Let X be a topological space, we say that a subset $A \subseteq X$ is a retract of X if there exists a extension $r : X \rightarrow A$. i.e. a continuous map with $r(a) = a, a \in A$

$$r|_A = id_A$$

Special case of homotopy:

Definition 6.1.1.2

1. A **deformation retraction** from X to A is a homotopy $F : X \times [0, 1] \rightarrow X$ which has following three properties:

- $F_0 = id_X$;
- F is a homotopy rel A , i.e. for all $a \in A$ and all $t \in [0, 1]$ we have

$$F(a, t) = a$$

- $F_1(X) = A$;

2. We say that A is a **deformation retract** of X if there exists a deformation retraction from X to A . We say a point $a \in A$ is a **deformation retract** of X if the subset $\{a\}$ is a deformation retract of X .
3. We say that a restriction $r : X \rightarrow A$ is realized by a deformation retraction if there exists a deformation retraction F from X to A with $F_1 = r$.

6.1.2 Joins, Mapping Cones, Cylinders and Tori

6.1.3 Cofibrations

6.2 Relative Homotopy Groups

6.3 Fibration

6.3.1 Definitions

Definition 6.3.1.1 Let $p : Y \rightarrow B$ be a continuous map between topological spaces:

- Let X be a topological space. We say that the map $p : Y \rightarrow B$ has the **homotopy lifting property with respect** to X if given any continuous map $F : X \times [0, 1] \rightarrow B$ and given any continuous map $\hat{f} : X \times \{0\} \rightarrow Y$ with $p \circ \hat{f} = F \circ i$ there exists a continuous map $\hat{F} : X \times [0, 1] \rightarrow Y$ with following diagram commutes

$$\begin{array}{ccc}
 X \times \{0\} & \xrightarrow{\forall \hat{f}} & Y \\
 \downarrow i & \nearrow \exists \hat{F} & \downarrow p \\
 X \times [0, 1] & \xrightarrow{\forall F} & B
 \end{array}$$

- We say the map $p : Y \rightarrow B$ is a **fibration** if p has the homotopy lifting property with respect to all topological space X .

Remark Sometimes called **Hurewicz fibration**

- We say the map $p : Y \rightarrow B$ is a **Serre fibration** if p has the homotopy lifting property with respect to all closed balls $\bar{B}^n, n \in \mathbb{N}$, so Serre fibration is not necessarily a fibration.

6.4 Homotopy of Fibre Bundles

These contents will be given in fibre bundles.

6.5 Homotopy Groups of Lie Groups

6.6 Thom-Pontryagin Theory

6.7 Homotopy Group of Sphere

Chapter 7 The Model Structure

7.1 Homotopical Algebra

7.1.1 Closed Model Category

Definition 7.1.1.1 A closed model category $\mathcal{C}$ is a category which is equipped with three classes of morphisms:

- Coribrations;
- Fibrations;
- Weak Equivalences;

Together with following lemmas:

- (CM1) The category $\mathcal{C}$ is closed under all finite limits and colimits;
- (CM2) Suppose that the following diagram commutes in $\mathcal{C}$:

$$\begin{array}{ccc} X & \xrightarrow{\quad} & Y \\ & \searrow & \swarrow \\ & Z & \end{array}$$

- (CM3) If f retract of g and g is weak equivalence, fibration or cofibration, then so is f ;
- (CM4) Suppose that we are given a commutative solid arrow diagram

$$\begin{array}{ccc} U & \xrightarrow{\quad} & X \\ \downarrow & \nearrow & \downarrow p \\ V & \xrightarrow{\quad} & Y \end{array}$$

- (CM5) Any map $f : X \rightarrow Y$ may be factored
 - $f = p \circ i$ where: p is fibration; i is trivial cofibration;
 - $f = q \circ j$ where: q is a trivial cofibration, j is a cofibration.

Especially: A map is a **trivial fibration (acyclic fibration)** if it is both a fibration and a weak equivalence; Dually, a trivial cofibration is a map which is simultaneously a cofibration and a equivalence;

◇ **Lemma 7.1.1.2** Suppose $\mathcal{C}$ be a closed model category, then we have the following

-
-
-
-

◦ **Proof ...**

□

★ **Corollary 7.1.1.3**

- The classes cofibrations and
-

◦ **Proof ...**

□

Cylinder Object

Definition 7.1.1.4 A **cylinder object** for an object A in closed model category C to be the commutative triangle

7.2 Simplicial Categories

Chapter 8 The Classical Results and Constructions

8.1 Homotopy Fibre Sequence

Chapter 9 Abstract Homotopy Theory

9.1 Weak Homotopy Equivalences

Definition 9.1.0.1 A category with weak equivalences is

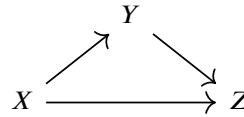
- A category C ;
- A subclass $W \subseteq Mor(C)$ of its morphisms

Such that:

- W contains all the isomorphisms of C ;

$$Isom(C) \subseteq W$$

- W is closed under **two out of three**: In every commuting diagram in C of the form



if two of the three morphisms are in W , then so is the third.

Remark With the weak equivalence category (C, W) , we can construct a new homotopy theory (via simplicial method)

$$L(C, W)$$

9.1.1 Model Category

Definition 9.1.1.1 A model category is

- A category C with all limits and colimits;
- Three subclasses $W, Fib, Cof \subseteq Mor(C)$ of its morphisms;

such that

- The class W make C into a category with weak equivalences;
- The pairs $(W \cap Cof, Fib)$ and $(Cof, W \cap Fib)$ are both weak factorization systems

One says:

- Elements in W are weak equivalences;
- Elements in Cof are **cofibrations**;
- Elements in Fib are **fibrations**;
- Elements in $W \cap Cof$ are **acyclic cofibrations**;
- Elements in $W \cap Fib$ are **acyclic fibrations**;

9.1.2 Factorization System

Definition 9.1.2.1 Let $\mathcal{C}$ be any category.

- Given a diagram in $\mathcal{C}$ of the form

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow p & & \\ B & & \end{array}$$

then, an **extension** of the morphism f along the morphism p is a completion of a commuting diagram of the form

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow p & \nearrow \bar{p} & \\ B & & \end{array}$$

- Dually, given a diagram of the form

$$\begin{array}{ccc} & & A \\ & \nearrow \bar{f} & \downarrow p \\ X & \xrightarrow{f} & Y \end{array}$$

is a **lift**.

- Consider the combining: Given a commuting square

$$\begin{array}{ccc} X_1 & \longrightarrow & Y_1 \\ \downarrow & & \downarrow \\ X_2 & \longrightarrow & Y_2 \end{array}$$

a **lifting** in the diagram is a completion to a commuting diagram of the form:

$$\begin{array}{ccc} X_1 & \xrightarrow{f_1} & Y_1 \\ \downarrow p_l & \nearrow & \downarrow p_r \\ X_2 & \xrightarrow{f_2} & Y_2 \end{array}$$

Given a sub-class of morphisms $K \subseteq \text{Mor}(\mathcal{C})$, then

-
-

Weak Factorization System

Definition 9.1.2.2 A **weak factorization system (WFS)** on a category $\mathcal{C}$ is a pair of classes of morphisms of $\mathcal{C}$ such that

-
-

Functorial Factorization

Definition 9.1.2.3 For $\mathcal{C}$ a category, a **functorial factorization** of the morphisms in $\mathcal{C}$ a functor

$$fact : \mathcal{C}^{\Delta[1]} \rightarrow \mathcal{C}^{\Delta[2]}$$

which is a section of the composition functor $d_1 : \mathcal{C}^{\Delta[2]} \rightarrow \mathcal{C}^{\Delta[1]}$.

9.2 Homotopy

Definition 9.2.0.1 Let $\mathcal{C}$ be a model category, and $X \in \mathcal{C}$ be an object:

- A **path space** $Path(X)$ is a factorization of the diagonal $\Delta_X : X \rightarrow X \times X$ as

$$\Delta_X : X \xrightarrow[\in W]{i} Path(X) \xrightarrow[\in Fib]{(p_0, p_1)} X \times X$$

where $i : X \rightarrow Path(X)$ is a weak equivalence and $(p_0, p_1) : Path(X) \rightarrow X \times X$ is a fibration;

- A **cylinder object** $Cyl(X)$ for X is a factorization of codiagonal $\nabla_X : X \sqcup X \rightarrow Cyl(X)$ is a cofibration.
-
-

9.3 The Homotopy Category

Chapter 10 Singular Cohomology

10.1 The Essentials

10.1.1 The Idea behind Singular Homology

Definition 10.1.1.1 Define:

- We say a set of points of $\mathbb{R}^d$ are in **general position** if for no k of them lies in $k - 2$ dimensional flat, $k = 2, \dots, d + 1$.
Especially, for $k + 1$ points, they are not contained in $(k - 1)$ dimensional affine flat. $\dim(\text{span}(P_i - P_0)) = k$.
- Given any finite subset $S \subseteq \mathbb{R}^n$, denote

$$\text{Hull}(S) := \left\{ \sum_{x \in S} \lambda_x \cdot x \mid \lambda_x \in \mathbb{R}_{\geq 0}, \sum \lambda_x = 1 \right\}$$

The **convex hull** of S ;

- A **k-simplex** in $\mathbb{R}^n$ is the convex hull of $k + 1$ elements that are in general position.
- Let $\sigma \subseteq \mathbb{R}^n$ be a k -simplex:
 - There exists a unique $V \subseteq \mathbb{R}^n$ with $k + 1$ and with $\sigma = \text{Hull}(V)$, we refer to V as the vertex set of σ .
 - Let V be the vertex of σ , for any subset $T \subseteq V$ with $(l + 1)$ elements we refer to $\text{Hull}(T)$ as an l -face of $\sigma = \text{Hull}(V)$.

Then

Definition 10.1.1.2 A **simplicial complex** in $\mathbb{R}^n$ is a finite family $X = \{\sigma_i\}$ of simplices in $\mathbb{R}^n$ with following properties:

- Any face of a simplex is also contained in the family; i.e. for $T \subseteq V$, $\text{Hull}(T) \in X$.
- The intersection of any two simplices is either empty or it is a single simplex, which is a face of both simplices.

Given a simplicial complex $X = \{\sigma_i\}_{i=1, \dots, k}$ as above we denote by X also the subspace

$$\sigma_1 \cup \dots \cup \sigma_k \subseteq \mathbb{R}^n$$

Chain, Boundary

Definition 10.1.1.3 Let X be a simplicial complex:

- An n -chain in X is a collection of n -simplices of X . In other words: It is a subset of the set of n -simplices of X ;
- The boundary $\partial\sigma$ of an n -chain σ is the union of all $(n - 1)$ simplices that are faces of an odd number n -simplices of the n -chain;
- We say an n -chain is an n -cycle if its boundary is empty set of $(n - 1)$ -simplices.

Remark Why odd? Consider a pair of triangles, omitted.

10.2 The Singular Homology Groups of a Topological Space

10.2.1 Δ -Sets

Definition 10.2.1.1 Let $n \in \mathbb{N}$:

- Given $i \in \{0, \dots, n\}$ we write $v_i := (0, \dots, 0, 1, 0, \dots, 0)$;
- Given $n \in \mathbb{N}$ we refer to the convex hull of $v_0, \dots, v_n \in \mathbb{R}^{n+1}$

$$\Delta^n := \{(x_0, \dots, x_n) \in \mathbb{R}^{n+1} \mid \sum x_i = 1, x_i \geq 0, i = 0, \dots, n\}$$

as the standard n -simplex;

- Given the standard n -simplex Δ^n we define the boundary as

$$\partial\Delta^n := \cup_{i=0}^n \{(t_0, \dots, t_n) \in \Delta^n \mid t_i = 0\}$$

- We write

$$\mathring{\Delta}^n := \Delta^n - \partial\Delta^n$$

It is equipped with a smooth manifold structure.

◇ **Lemma 10.2.1.2** (Δ^n -Homeomorphism Lemma) Let $n \in \mathbb{N}$,

- we set

$$\triangleleft_n := \{(x_1, \dots, x_n) \in [0, 1]^n \mid \sum x_i \in [0, 1]\}$$

we refer to it as the planar n -simplex. The maps

$$\begin{aligned} \Phi : \Delta^n &\rightarrow \triangleleft_n & \Psi : \triangleleft_n &\rightarrow \Delta^n \\ (t_0, \dots, t_n) &\mapsto (t_0, \dots, t_{n-1}) & (x_1, \dots, x_n) &\mapsto (x_1, \dots, x_n, 1 - \sum x_i) \end{aligned}$$

are homeomorphisms that are inverses of one another;

- The map

$$\Delta_n \longrightarrow \bar{B}^n$$

Apply the homeomorphism on $\triangleleft_n$ and $Q = (\frac{1}{n+1}, \dots, \frac{1}{n+1})$.

◦ **Proof** Omitted. □

10.2.2 The Singular Chain Group

Definition 10.2.2.1

- Let X be a topological space, $n \in \mathbb{N}$. A **singular** n -simplex is a continuous map

$$\sigma : \Delta^n \rightarrow X$$

-
-

•

10.2.3 Singular Homology Groups

First Calculaton of Homology Groups

Singular 1-Chains

10.3 Homology and Homotopies

10.3.1 Chain Homotopies

Omitted

10.3.2 Homology and Homotopic Maps

♠ **Proposition 10.3.2.1** (Homotopy- H_* -Proposition) Let X, Y be two topological spaces, $f, g : X \rightarrow Y$ be two continuous maps.

If f and g are homotopic, then the following hold:

-
-

◦ **Proof ...**

□

★ **Corollary 10.3.2.2** (Homotopy Category H_* Functor Corollary)

◦ **Proof ...**

□

★ **Corollary 10.3.2.3** (Homotopy Equivalence H_* Corollary)

◦ **Proof ...**

□

★ **Corollary 10.3.2.4** (Contractible H_* Corollary)

◦ **Proof ...**

□

10.4 Exact Sequences and Long Exact Sequences

10.5 Relative Homology

10.6 Reduced Homology

10.7 The Excision Theorem

10.7.1 The Excision Theorem

► **Theorem** Let X be a topological space, and let

$$Z \subseteq A \subseteq X$$

be subsets. If

$$\bar{Z} \subseteq A$$

Then, the inclusion

$$(X - Z, A - Z) \rightarrow (X, A) \Rightarrow H_n(X - Z, A - Z) \xrightarrow{\cong} H_n(X, A)$$

Definition 10.7.1.1 Define:

- Define the **cover** of X by

$$X = \cup_i U_i$$

where $U_i \subseteq X$.

- A cover $\{U_i\}$ is **comfortable** if

$$X = \cup_i \dot{U}_i$$

Definition 10.7.1.2 Let X be a topological space. Given a **comfortable** we consider the following subgroup of $C_n(X)$:

$$C_n^{\mathcal{U}}(X) := \{ \sum a_j \cdot \sigma_j \mid \text{for each } j, \text{ there exists a } U_i \in \mathcal{U} \text{ such that the image of } \sigma_j \text{ lies in } U_i \}$$

with $H_n^{\mathcal{U}}(X) := \ker(\partial_n) / \text{im}(\partial_{n+1})$ is defined by singular chains which are 'small' in the sense that the image of each singular simplex lies in some U_i .

This is a subcomplex:

...

♠ **Proposition 10.7.1.3** (Comfortable Cover Proposition) Let X be a topological space, let $\mathcal{U} = \{U_i\}$ be a comfortable cover of X . The inclusion map

...

induces for every $n \in \mathbb{N}$ an isomorphism

$$H_n^{\mathcal{U}}(X) \xrightarrow{\cong} C_*(X)$$

◦ **Proof**

□

♠ **Proposition 10.7.1.4** (Subdivision-Natural Lemma)

◦ **Proof ...**

□

◇ **Lemma 10.7.1.5** (Chain Subdivision Lemma) Let X be a topological space. The maps

$$u_n : C_n(X) \rightarrow C_n(X)$$

for a chain map from the chain complex $(C_*(X), \partial)$ to itself.

◦ **Proof ...** □

◇ **Lemma 10.7.1.6** (Subdivision-Chain Homotopy Lemma) Let X be a topological space. The chain maps

$$u_n : C_n(X) \rightarrow C_n(X)$$

are chain homotopic to the identity.

◦ **Proof ...** □

◦ **Proof ...** □

► **Theorem** (Excision Theorem) Let X be a topological space, and let K be a subset of X , $U \supseteq \bar{K}$. The inclusion

$$(U, U - K) \rightarrow (X, X - K) \Rightarrow H_n(U, U - K) \xrightarrow{\cong} H_n(X, X - K)$$

◦ **Proof** $H_n(U, U - K) = H_n(X - Z, A - Z)$: Let $A = X - K$; $Z = X - U$.

Then we have

$$H_n(X - Z, A - Z) \xrightarrow{\cong} H_n(X, A) = H_n(X, X - K)$$

□

♣ **Example 10.7.1.7** ...

Excision of Manifold

► **Theorem** (Manifold-Excision Theorem) Let X be an m -dimensional topological manifold, $A, B \subseteq X$ be two codimension-zero submanifolds such that the following statements hold:

- $X = A \cup B$;
- $A \cap B$ is the union of components of ∂A ;
- A, B are closed subsets of X .

For every $n \in \mathbb{N}$, the **inclusion-induced map**

$$H_n(B, A \cap B) \rightarrow H_n(X, A)$$

is an isomorphism.

◦ **Proof ...** □

CW-Excision

► **Theorem** (CW-Excision Theorem) Let X be a CW-complex, and let A, B be two subcomplexes of X with

...

Excision-via-Deformation Theorem

► **Theorem** (Excision via Deformation Theorem)

◦ **Proof** ...

□

10.7.2 Local Homology and Explicit Generators of Homology Groups

10.8 Local Homology

10.9 The Mayer-Vietoris Sequence

10.10 Homology Groups and Direct Colimits

10.11 Cellular Homology

10.11.1 Relative Homology Groups of CW-complexes

Notation Let X be a CW-complex:

-
-

◊ **Lemma 10.11.1.1** (Skeleton- H_* -Lemma) Let X be a CW-complex and $n \in \mathbb{N}$:

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-
-
-

◦ **Proof** ...

□

10.12 Euler Characteristic

10.12.1 The Euler Characteristic of a CW-complex

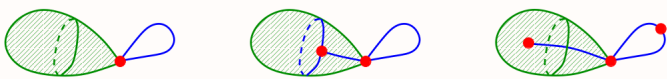
Definition 10.12.1.1 Given a finite CW-complex X , we define its **Euler characteristic** $\chi(X)$ as follows

$$\chi(X) := \sum_{n \in \mathbb{N}} (-1)^n \cdot |Cell_n(X)|$$

♣ **Example 10.12.1.2** ...

♣ **Example 10.12.1.3** (Euler Characteristic of $S^2 \vee S^1$) The table shows the stability of Euler-characteristic, in different CW-decomposition:

In the figure below we show the topological space $S^2 \vee S^1$ with three different CW-structures.



number of 0-cells	1	2	3
number of 1-cells	1	3	3
number of 2-cells	1	2	1
Euler characteristic	1	1	1

10.12.2 Betti Numbers

10.12.3 Relative Characteristic

◇ **Lemma 10.12.3.1** ...

◦ **Proof** ...

□

10.12.4 Properties

Union

Quotient

Product

Covering

♠ **Proposition 10.12.4.1** Let $p : \tilde{X} \rightarrow X$ be a finite covering of a finite path-connected non-empty CW-complex X . We equip $\tilde{X}$ with the structure of a CW-complex given by the CW-complex covering proposition, then

$$\chi(\tilde{X}) = [\tilde{X} : X] \cdot \chi(X)$$

◦ **Proof** ...

□

10.12.5 Calculations of Euler Characteristics

10.12.6 Applications

10.13 The Fundamental Theorem of HA

10.14 The Eilenberg-Zilber Theorem and the Künneth Theorem

Chapter 11 Homology

According to **Dold-Kan correspondence** we have

$$Ab^{\Delta_{op}} \Leftrightarrow Ch_{\geq 0}(AbD)$$

The **homology groups** of a chain complex of abelian groups are the image under the identification of the homotopy groups and the corresponding ∞ -groupoids.

Thus, we have the following slogan

Homology=Homotopy, under Dold-Kan Correspondence

11.1 Eilenberg-Steenrod Axiom

† **Axiom** (Eilenberg-Steenrod) A homology theory $H_* = (H_*, \partial_*)$ with values in R -modules is a covariant functor

$$H_* : Top^2 \rightarrow \mathbb{Z} - grad \quad R - Mod$$

Together with a natural transformation $\partial : H_i(X, A) \rightarrow H_{i-1}(A, \emptyset)$ called the **boundary map** by

$$\partial_i(X, A) : H_i(X, A) \rightarrow H_{i-1} \circ I(X, A) \quad I : (X, A) \mapsto (A, \emptyset)$$

1. Homotopy:
2. Excision:
3. Dimension:
4. Additivity:
5. Exactness:

11.1.1

◇ **Lemma 11.1.1.1** (Exact Sequence of Triples)

◦ **Proof ...**

□

► **Theorem** (Mayer-Vietoris Sequence) Suppose (X, X_1, X_2) be a

◦ **Proof ...**

□

★ **Corollary 11.1.1.2** (Mayer-Vietoris Sequence for Pushout)

Chapter 12 Singular Homology

12.1 Fundamental

12.1.1 Simplicial Chain Group, Simplicial Homology Group

Simplicial Chain Group

Definition 12.1.1.1 Let X be a simplicial complex

- We define the n -th simplicial chain group as

$$C_n^{simp}(X) := \{\text{Set of all } n\text{-chains}\} = \mathbb{Z}_2^{\text{set of } n\text{-simplices of } X}$$

- Define the boundary map

$$\begin{aligned} \partial_n : C_n^{simp}(X) &\rightarrow C_{n-1}^{simp}(X) \\ \sum a_i \cdot \sigma_i &\mapsto \sum a_i \cdot \partial \sigma_i \end{aligned}$$

- Sequence:

$$\dots \xrightarrow{\partial_3} C_2^{simp}(X) \xrightarrow{\partial_2} C_1^{simp}(X) \xrightarrow{\partial_1} C_0^{simp}(X) \rightarrow 0$$

which is the **simplicial chain complex** of X .

◇ **Lemma 12.1.1.2** (Simplicial-Double Boundary Lemma) Let X be a simplicial complex. For every $n \in \mathbb{N}_0$, the composition

$$0 = \partial_n \circ \partial_{n+1} : C_{n+1}^{simp}(X) \rightarrow C_{n-1}^{simp}(X)$$

- **Proof** Suppose σ be an $(n+1)$ -simplex with vertex set $V = \{x_0, \dots, x_{n+1}\}$. We have

$$\begin{aligned} \partial_n(\partial_{n+1}(\sigma)) &= \partial_n\left(\sum_{i=0}^{n+1} \text{Hull}(\{x_0, \dots, \hat{x}_i, \dots, x_{n+1}\})\right) \\ &= \sum_{i=0}^{n+1} \sum_{j=0, i \neq j}^{n+1} \text{Hull}(\{x_0, \dots, \hat{x}_i, \dots, \hat{x}_j, \dots, x_{n+1}\}) \\ &= 0 \end{aligned}$$

Because: Each Hull appears twice. □

Simplicial Homology Group

Definition 12.1.1.3 Let X be a simplicial complex. Furthermore let $n \in \mathbb{N}$, it follows immediately from above, we have the inclusion

$$\text{im}(\partial_{n+1}^{simp}(X) \rightarrow C_n^{simp}(X)) \subseteq \ker(\partial_n : C_n^{simp}(X) \rightarrow C_{n-1}^{simp}(X))$$

Thus, define the n -th simplicial homology group as follows:

$$H_n^{simp}(X) : Ker(\partial_n)/Im(\partial_{n+1})$$

12.1.2 Examples

Skip. <https://friedl.app.uni-regensburg.de/papers/1t-total-public-october-7-2024.pdf> P2244

12.2 The Singular Homology Groups of a Topological Space

12.2.1 Review: Algebraic Complex

In HA.

- Complex;
- Homological Functor;
- Operation of Complexes.

12.2.2 The Singular Chain Complex

12.2.3 Singular n -Simplex

Definition 12.2.3.1

- Let X be a topological space and $n \in \mathbb{N}$. A **singular n -simplex** is a continuous map

$$\sigma : \Delta^n \rightarrow X$$

- Define the n -th singular chain group as follows:

$$C_n(X) := \mathbb{Z}(\{\sigma \mid \sigma \text{ is singular } n\text{-simplices of } X\})$$

- We refer to an element in $C_n(X)$ as a singular n -chain in X ;

Face Map

Definition 12.2.3.2 Let $n \in \mathbb{N}$. For $j \in \{0, \dots, n\}$ we define the j -th face map as follows

$$\begin{aligned} i_j^n : \Delta^{n-1} &\rightarrow \Delta^n \\ (t, \dots, t_{n-1}) &\mapsto (t_0, \dots, t_{j-1}, 0, \dots, t_{n-1}) \end{aligned}$$

Boundary

Definition 12.2.3.3 Let X be a topological space and let $\sigma : \Delta^n \rightarrow X$ be a singular n -simplex. The **boundary** $\partial_n \sigma \in C_{n-1}(X)$ is defined as

$$\partial_n \sigma := \sum_{j=0}^n (-1)^j \cdot \sigma \circ i_j^n$$

Then, the map extends uniquely to a linear map

$$\begin{aligned}\partial_n : C_n(X) &\rightarrow C_{n-1}(X) \\ \sum a_r \cdot \sigma_r &\mapsto \sum a_r \cdot \partial \sigma_r\end{aligned}$$

♠ **Proposition 12.2.3.4** Let X be a topological space, then the composition

$$\partial_{n-1} \circ \partial_n : C_n(X) \rightarrow C_{n-2}(X)$$

is the zero map.

◦ **Proof** We have

$$\begin{aligned}\partial_{n-1}(\partial_n(\sigma)) &= \partial_{n-1}\left(\sum_{j=0}^n (-1)^j \cdot \sigma \circ i_j^n\right) \\ &= \sum_{k=0}^{n-1} \cdot \sum_{j=0}^n (\sigma \circ i_j^n) \circ i_k^{n-1} \\ &= \sum_{0=k < j \leq n} (-1)^{k+j} \cdot \sigma \circ (i_j^n \circ i_k^{n-1}) + \sum_{0=j \leq k \leq n-1} (-1)^{k+j} \cdot \sigma \circ (i_j^n \circ i_k^{n-1}) \\ &= \sum_{0=k < j \leq n} (-1)^{i+j} \cdot \sigma \circ (i_j^n \circ i_k^{n-1}) + \sum_{0 \leq j \leq k \leq n-1} (-1)^{k+j} \cdot \sigma \circ (i_{k+1}^n \circ i_j^{n-1}) \quad i_j^n \circ i_k^{n-1} + k = i_{k+1}^n \circ i_j^{n-1} \\ &= \sum_{0=k < j \leq n} (-1)^{i+j} \cdot \sigma \circ (i_j^n \circ i_k^{n-1}) + \sum_{0 \leq j \leq k \leq n-1} (-1)^{k+j} \cdot \sigma \circ (i_{k+1}^n \circ i_j^{n-1}) \\ &= 0\end{aligned}$$

□

Morphisms

Definition 12.2.3.5 Let $f : X \rightarrow Y$ be a continuous map between topological spaces. For each $n \in \mathbb{N}$ we refer to

...

◇ **Lemma 12.2.3.6** (C_* -Functor Lemma)

◦ **Proof** ...

□

Path Components

◇ **Lemma 12.2.3.7** (Path-Component C_* -Lemma) Let X be a topological space with path components $\{X_a\}_{a \in A}$. Given $a \in A$ we denote by $i_a : X_a \hookrightarrow X$ the inclusion map. The map

$$\bigoplus_{a \in A} i_a : \bigoplus_{a \in A} C_*(X_a) \rightarrow C_*(X)$$

is an isomorphism of chain complexes.

◦ **Proof** ...

□

◇ Lemma 12.2.3.8 ...

◦ Proof ...

□

Singular Homology Groups

Definition 12.2.3.9 Let X

Definition 12.2.3.10 Let X be a topological space:

-
-
-

Functor $H_n(-)$

◇ Lemma 12.2.3.11

- Let $f : X \rightarrow Y$ be a continuous map between topological spaces. The map

...

- $\mathcal{T}op \rightarrow AbGroup$ defined as followings:

–
–

◦ Proof ...

□

12.2.4 Calculations of Homology Groups

Empty Spaces

Augment Lemma

Singular 1-Chain

12.3 Homology and Homotopies

Recall the definitions:

- Homotopy, Homotopy Equivalence;
- Chain Homotopy;

12.3.1 Homotopy and Homology Maps

♠ Proposition 12.3.1.1 (Homotopy- H_* -Proposition)

◦ Proof ...

□

★ Corollary 12.3.1.2 ...

◦ Proof ...

□

12.4 Exact Sequence, Long Exact Sequences

12.5 Relative Homology

Definition 12.5.0.1 Let (X, A) be a pair of topological spaces. We view each chain group $C_n(A)$ as a subgroup of $C_n(X)$. Thus we can define

$$C_n(X, A) := C_n(X)/C_n(A)$$

and consider the boundary map

$$\begin{aligned} \partial_n : C_n(X, A) &\rightarrow C_{n-1}(X, A) \\ [c] &\mapsto [\partial_n c] \end{aligned}$$

We refer to

$$\dots \xrightarrow{\partial_{n+2}} C_{n+1}(X, A) \xrightarrow{\partial_{n+1}} C_n(X, A) \xrightarrow{\partial_n} C_{n-1}(X, A) \xrightarrow{\partial_{n-1}} \dots$$

And

$$H_n(X, A) := H_n(C_*(X, A))$$

Functoriality

Firstly, take the notion

$$\begin{aligned} Ob(PairTop) &:= \text{all pairs of topological spaces} \\ Mor((X, A), (Y, B)) &:= \{\text{all continuous maps } f : X \rightarrow Y, f(A) \subseteq B\} \end{aligned}$$

Definition 12.5.0.2 Let $f : (X, A) \rightarrow (Y, B)$ be a continuous map of pairs of topological spaces. i.e. a continuous map

$$f : X \rightarrow Y \quad f(A) \subseteq B$$

It follows that: There exists a chain map

$$\begin{aligned} f_* : C_n(X, A) &\rightarrow C_n(Y, B) \\ H_n(f_*) : H_n(X, A) &\rightarrow H_n(Y, B) \end{aligned}$$

◇ **Lemma 12.5.0.3** (Relative- H_* -Functor Lemma) The maps

$$\begin{aligned} (X, A) &\mapsto (C_*(X, A), \partial_*) \\ (f : (X, A) \rightarrow (Y, B)) &\mapsto (f_* : C_*(X, A) \rightarrow C_*(Y, B)) \end{aligned}$$

define a covariant functor from the subcategory $PairTop$ of pairs of topological spaces to the category $ChCplx$ of chain complexes and for each $n \in \mathbb{N}$,

$$\begin{aligned} (X, A) &\rightarrow H_n(X, A) \\ (f : (X, A) \rightarrow (Y, B)) &\mapsto (f_* : H_n(X, A) \rightarrow H_n(Y, B)) \end{aligned}$$

- **Proof** Immediately from the definition. □

12.5.1 Long Exact Sequence of a Triple

The Triple of Topological spaces

Definition 12.5.1.1

- A **triple of topological space** is a triple (X, B, A) where X is a topological space and $A \subseteq B \subseteq X$ are subspaces;
- Morphisms $f : (X, B, A) \rightarrow (X', B', A')$ such that

$$f : X \rightarrow X', f(B) \subseteq B', f(A) \subseteq A'$$

♠ Proposition 12.5.1.2 (LES of Triple Propositions)

- **Proof** ... □

12.5.2 Relative Homology Groups and Homotopies

♠ **Proposition 12.5.2.1** Let $f, g : (X, A) \rightarrow (Y, B)$ be two continuous maps between pairs of topological spaces. If f, g are homotopic, then the chain maps

$$f_*, g_* : C_*(X, A) \rightarrow C_*(Y, B)$$

are chain homotopic. In particular for every $n \in \mathbb{N}$ we have

$$f_* = g_* : H_n(X, A) \rightarrow H_n(Y, B)$$

- **Proof** ... □

♠ **Proposition 12.5.2.2** (LES-of-Triple Proposition) Let X be a topological space and let $A \subseteq B \subseteq X$ be two subspaces. Let $i : (B, A) \rightarrow (X, A)$ and by $p : (X, A) \rightarrow (X, B)$ be the obvious maps

- For each $n \in \mathbb{N}$, the following map is a well-defined group homomorphism

$$\partial_n : H_n(X, B) \rightarrow H_{n-1}(B, A)$$

$$[\sum s_j \cdot (\sigma_j : \Delta^n \rightarrow X)] \mapsto [\sum_{j=1}^k s_j \cdot \partial \sigma_j]$$

We refer ∂_n as a **connecting homomorphism** of the triple (X, B, A) .

- The connecting homomorphisms are natural: In particular, for any continuous map

$$f : (X, A, B) \rightarrow (\hat{X}, \hat{A}, \hat{B})$$

of triples of topological spaces the following diagram commutes:

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & H_n(B, A) & \longrightarrow & H_n(X, A) & \longrightarrow & H_n(X, B) \longrightarrow H_{n-1}(B, A) \longrightarrow \cdots \\
& & \downarrow & & \downarrow & & \downarrow \\
\cdots & \longrightarrow & H_n(\hat{B}, \hat{A}) & \longrightarrow & H_n(\hat{X}, \hat{A}) & \longrightarrow & H_n(\hat{X}, \hat{B}) \longrightarrow H_{n-1}(\hat{B}, \hat{A}) \longrightarrow \cdots
\end{array}$$

- The maps

$$\cdots \longrightarrow H_n(B, A) \longrightarrow H_n(X, A) \longrightarrow H_n(X, B) \longrightarrow H_{n-1}(B, A) \longrightarrow \cdots$$

form an exact sequence, called **long exact sequence** of the triple (X, B, A) .

- **Proof** Firstly, we should prove the exactness of

$$0 \longrightarrow C_*(B, A) \longrightarrow C_*(X, A) \longrightarrow C_*(X, B) \longrightarrow 0$$

Immedaitey from the definition.

Then: Suppose $z \in H_n(X, B)$ with $z = [\sum s_j \cdot (\sigma_j : \Delta^n \rightarrow X)]$

Or: Notice that (depends on snake lemma) we have

$$0 \rightarrow \text{Ker}(d_n^{C_*(B,A)}) \rightarrow \text{Ker}(d_n^{C_*(X,A)}) \rightarrow \text{Ker}(d_n^{C_*(X,B)}) \xrightarrow{\delta} \text{Coker}(d_n^{C_*(B,A)}) \rightarrow \text{Coker}(d_n^{C_*(X,A)}) \rightarrow \text{Coker}(d_n^{C_*(X,B)}) \rightarrow 0$$

Then, for δ , consider the following square

$$\begin{array}{ccccccc}
& & \text{Ker}(d_n^{C_*(B,A)}) & \longrightarrow & \text{Ker}(d_n^{C_*(X,A)}) & \longrightarrow & \text{Ker}(d_n^{C_*(X,B)}) \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & C_n(B, A) & \longrightarrow & C_n(X, A) & \longrightarrow & C_n(X, B) \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & C_{n-1}(B, A) & \longrightarrow & C_{n-1}(X, A) & \longrightarrow & C_{n-1}(X, B) \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
& & \text{Coker}(d_n^{C_*(B,A)}) & \longrightarrow & \text{Coker}(d_n^{C_*(X,A)}) & \longrightarrow & \text{Ker}(d_n^{C_*(X,B)})
\end{array}$$

Diagram chasing: Suppose $c \in \text{Ker}(d_n^{C_*(X,B)}) \subseteq C_n(X, B)$. We have

$$\begin{aligned}
p_*(b) &=, b \in \text{Ker}(d_n^{C_n(X,A)}) \subseteq C_n(X, A) \\
i_*(a) &= d(b)
\end{aligned}$$

And notice that $a \in \text{ker}(d_{n-1})$. Because of $C_n(B, A) \rightarrow C_n(X, A) \rightarrow C_{n-1}(X, A)$ maps a to 0. Now consider the square $C_{n-1}(B, A) \rightarrow C_{n-2}(B, A) \rightarrow C_{n-2}(X, A)$. $\square$

♠ Proposition 12.5.2.3 (LES-of-Pair Proposition)

- Let X be a topological space and $B \subseteq X$ be a subspace. The connecting homomorphism of the LES-of-Triple proposition applied to $A = \emptyset$ gives rise to an exact seugence

...

- The connecting homomorphism is natural. In particular, given a continuous map

$$f : (X, A) \rightarrow (Y, B)$$

between pairs of topological spaces we obtain a commutative

$$H_n(A)$$

◦ **Proof ...** □

★ **Corollary 12.5.2.4** (Homotopy Equivalence-Relative- H_* -Corollary)

◦ **Proof ...** □

★ **Corollary 12.5.2.5** (Deformation Retract-Relative H_* -Corollary) Let X be a topological space. If $A \subseteq X$ is a deformation retract of X , then

$$H_n(X, A) = 0, n \in \mathbb{N}$$

◦ **Proof ...** □

12.5.3 Relative Homology Groups and Homotopies

♠ **Proposition 12.5.3.1** Let $f, g : (X, A) \rightarrow (Y, B)$ be two continuous maps between pairs of topological spaces. If f, g homotopic, then the chain maps

$$f_*, g_* : C_*(X, A) \rightarrow C_*(Y, B)$$

are chain homotopic, in particular for every $n \in \mathbb{N}$ we have

$$f_* = g_* : H_n(X, A) \rightarrow H_n(Y, B)$$

Let $f, g : (X, A) \rightarrow (Y, B)$ be two continuous maps between pairs of topological spaces and let

...

◦ **Proof ...** □

12.5.4 Excision

♠ **Proposition 12.5.4.1** Let X be a topological space, $A \subseteq X$ open and $Z \subseteq A$ closed, with

$$Z \subseteq A$$

The inclusion $(X - Z, A - Z) \rightarrow (X, A)$ induces an epimorphism

$$H_1(X - Z, A - Z) \rightarrow H_1(X, A)$$

◦ **Proof ...** □

12.5.5 Retract

♠ **Proposition 12.5.5.1** Let (X, A) be a pair of topological spaces. If A is a retract of X , i.e. exists $f : X \rightarrow A$ such that

$$f \circ i_A = f|_A = id_A$$

then for each $n \in \mathbb{N}$, there exists an isomorphism

$$H_n(X) \cong H_n(A) \oplus H_n(X, A)$$

◦ **Proof ...** □

12.6 Reduced Homology

Definition 12.6.0.1 Let X be a topological space. Then we know that the following is a generalized chain complex

$$\dots \xrightarrow{\partial_2} C_1(X) \xrightarrow{\partial_1} C_0(X) \xrightarrow{\epsilon_X} \mathbb{Z} \rightarrow 0$$

By $\epsilon_X : \sum r_i \cdot \sigma_i \mapsto \sum r_i$

We introduce the following definitions:

- We refer to this chain complex as the augmented chain complex $\hat{C}_*(X)$ of X ;
- Given $n \in \mathbb{Z}_{\geq -1}$ we refer to

$$\hat{H}_n(X) := H_n(\hat{C}_*(X))$$

◇ **Lemma 12.6.0.2** ...

◦ **Proof ...** □

12.7 The Excision Theorem

► **Theorem** (Excision Theorem) Let X be a topological space and let $Z \subseteq A \subseteq X$ be subsets. If the closure of Z is contained in the interior of A then the inclusion

$$(X - Z, A - Z) \rightarrow (X, A)$$

induces for each $n \in \mathbb{N}$ an isomorphism

$$H_n(X - Z, A - Z) \rightarrow H_n(X, A)$$

◦ **Proof ...** □

► **Theorem** (The Excision Theorem) Let X be topological space, K be a subset of X and U be a neighborhood of the closure of K . Then, the inclusion $(U, U - K) \rightarrow (X, X - K)$ induces for each $n \in \mathbb{N}$ an isomorphism

$$H_n(U, U - K) \rightarrow H_n(X, X - K)$$

Suspensions and Homology Groups

Definition 12.7.0.1 Let X be a non-empty topological space;

- Define the **suspension** of X to be the topological space

$$\Sigma(X) := (X \times [-1, 1]) / (X \times \{-1\}) / (X \times \{1\})$$

- Define:

- North Pole $N = [X \times \{1\}]$;
- South Pole $S = [X \times \{-1\}]$;

- Set:

$$C_-X := \Sigma(X) - \{N\} \quad C_+X := \Sigma(X) - \{S\}$$

- Given a continuous map $f : X \rightarrow Y$ between topological spaces, then

$$\begin{aligned} \Sigma(f) : \Sigma(X) &\rightarrow \Sigma(Y) \\ [(x, t)] &\mapsto [(f(x), t)] \end{aligned}$$

♠ **Proposition 12.7.0.2** (Suspension- H_* -Proposition)

◦ **Proof ...**

□

♠ **Proposition 12.7.0.3** (Reflection-on-Suspension- H_* -Proposition) Let

$$\begin{aligned} \rho : \Sigma(X) &\rightarrow \Sigma(X) \\ [(x, t)] &\mapsto [(x, -t)] \end{aligned}$$

be the natural reflection. For any $k \in \mathbb{N}$, the induced map

$$\rho_k : \hat{H}_k(\Sigma(X)) \rightarrow \hat{H}_k(\Sigma(X))$$

given by multiplication by -1 .

◦ **Proof ...**

□

Homology Group of Sphere

Brouwer's Fixed Point Theorem

Homology Group of Wedge

The Long Exact Sequence for Quotients

12.8 Local Homology

12.8.1 Generator of $H_n(\Delta^n, \partial\Delta^n)$

◇ Lemma 12.8.1.1 ...

◦ Proof ...

□

12.8.2 Local Homology Groups of a Topological Space

Definition 12.8.2.1 ...

12.9 Homology and Self-Maps of Spheres

12.9.1 The Degree

Definition 12.9.1.1 Let $(G, +)$ be a group such that $G, \Phi : G \rightarrow G$ be a homomorphism.

Note that Φ is given by multiplication by a unique integer which we refer to as the degree $\deg(\Phi)$.

◇ Lemma 12.9.1.2 (Algebraic Degree Lemma) Let $(G, +)$ be a group that is isomorphic to $\mathbb{Z}$:

- Let $g \in G$ be any non-trivial element. Furthermore let $\Phi : G \rightarrow G$ be a homomorphism. The degree $\deg(\Phi)$ is uniquely determined by the quality

$$\Phi(g) = \deg(\Phi) \cdot g$$

- Let $\varphi : G \rightarrow H$ be an isomorphism and let $f : H \rightarrow H$ be a homomorphism. Then,

$$\deg(\varphi^{-1} \circ f \circ \varphi : G \rightarrow G) = \deg(f : H \rightarrow H)$$

◦ Proof ...

□

Sphere-Degree

◇ Lemma 12.9.1.3 (Sphere-Degree Lemma) Let $n \in \mathbb{N}$ and $f, g : S^n \rightarrow S^n$ be two continuous maps:

- $\deg(id_{S^n})$;
- If f is not surjective, then $\deg(f) = 0$;
- If f is homotopic to g , then $\deg(f) = \deg(g)$;
- $\deg(f \circ g) = \deg(f) \cdot \deg(g)$;
- If f is a self-homeomorphism of S^n , then $\deg * f^{-1} = \deg(f)$.

◦ **Proof ...** □

◇ **Lemma 12.9.1.4** (Degree-of-Power Lemma) Given $k \in \mathbb{Z}$ we have

...

◦ **Proof ...** □

◇ **Lemma 12.9.1.5** (Sphere-Degree-Matrix Lemma) Given any $A \in O(n+1)$ we have

...

◦ **Proof ...** □

Suspension

◇ **Lemma 12.9.1.6** ...

◦ **Proof ...** □

12.9.2 The Hairy Ball Theorem

► **Theorem** (Sphere-Degree Fixed Point Theorem) Let $f : S^n \rightarrow S^n$ be a continuous map. If

...

◦ **Proof ...** □

♠ **Proposition 12.9.2.1** ...

◦ **Proof ...** □

► **Theorem** hairyBTheor'me'm

◦ **Proof ...** □

12.9.3 Local Degrees of Self-Maps of Spheres

Definition 12.9.3.1 Let $f : M \rightarrow N$ be a continuous map between two oriented n -dimensional smooth manifolds, $x \in M - \partial M$.

If f is a local diffeomorphism, then define the **local degree** at point x :

...

12.10 Mayer-Vietoris Sequence

◇ **Lemma 12.10.0.1** (Brratt-Whitehead, Algebraic) Consider the following commutative diagram of homomorphisms between abelian groups

$$\begin{array}{ccccccccc} \cdots & & C'_n & & C_n & & C''_n & & C'_{n-1} & & \cdots \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \cdots & & D'_n & & D_n & & d''_N & & D'_{n-1} & & \cdots \end{array}$$

Then, define

$$\Delta_n := \partial_n \circ (f''_n)^{-1} \circ q_n : D_n \rightarrow C'_{n-1}$$

The following sequence is exact

$$\cdots \rightarrow C'_n \xrightarrow{(i_n, -f'_n)} C_n \oplus D'_n \xrightarrow{f_n \oplus j_n} D_n \xrightarrow{\Delta_n} C'_{n-1} \rightarrow \cdots$$

◦ **Proof ...**

□

► **Theorem** (Mayer-Vietoris) Let X be a topological space and let $A, B \subseteq X$ be subspaces such that one of the following three conditions holds:

- X is the union of the interiors of A and B . i.e.

$$X = \mathring{A} \cup \mathring{B}$$

- X is a CW-complex and A, B are subcomplexes with $X = A \cup B$;
- X is a topological manifold and $A, B \subseteq X$ are two codimension-zero manifolds such that the following conditions are satisfied:
 - $X = A \cup B$;
 - $A \cap B$ is a union of components of ∂A and it is a union of components of ∂B ;
 - A and B are closed subsets of X ;

Then, by excision theorem; the CW-excision theorem; Manifold excision theorem each induced by

$$i_{(A, A \cap B)}^{(X, B)} : H_A(A, A \cap B) \rightarrow H_n(X, B)$$

is an isomorphism. We refer to the homomorphism

$$H_n(X) \rightarrow H_n(X, B) \rightarrow H_n(A, A \cap B) \xrightarrow{\partial_n} H_{n-1}(A \cap B)$$

as the connecting homomorphism

$$\partial_n : H_n(X) \rightarrow H_{n-1}(A \cap B)$$

corresponding to (X, A, B) . The connecting homomorphism is natural in a suitable sense and with the notation the following sequence is exact:

$$\cdots \rightarrow H_n(A \cap B) \rightarrow H_n(A) \oplus H_n(B) \rightarrow H_n(X) \rightarrow H_{n-1}(A \cap B) \rightarrow \cdots$$

Totally analogous statement also hold if we replace the homology groups of $A \cap B, A, B, X$ by corresponding reduced homology groups.

◦ **Proof ...**

□

Another Proof:

...

Definition 12.10.0.2 The long exact sequence of above is called **Mayer-Vietoris Sequence of the decomposition** $X = A \cup B$.

12.10.1 Applications

Suspension

♠ **Proposition 12.10.1.1** (Suspension- H_* -Proposition) Given any topological space X and given any $k \in \mathbb{Z}$ we have a natural isomorphism

$$\Sigma_X : \hat{H}_{k+1}(\Sigma(X)) \xrightarrow{\cong} H_k(X)$$

◦ **Proof ...**

□

Mapping Torus

Definition 12.10.1.2 Let X be a topological space and $f : X \rightarrow X$ be a continuous map. We refer to

$$Tor(X, f) := (X \times [0, 1]) / (x, 0) \sim (f(x), 1)$$

Especially,

$$\begin{aligned} \Phi : Tor(X, id) &\rightarrow X \times S^1 \\ [(x, t)] &\mapsto (x, \exp(2\pi it)) \end{aligned}$$

is an isomorphism.

♣ **Example 12.10.1.3** (Möbius Band) $X = [-1, 1]$

♠ **Proposition 12.10.1.4** (Mapping Torus H_* -Proposition) Let X be a topological space, $f : X \rightarrow X$ be a continuous map. Then, denote

$$i : X \rightarrow Tor(X, f) \quad x \mapsto [(x, 0)]$$

for every $n \in \mathbb{N}$, there exists a natural homomorphism

$$\partial_n : H_n(\text{Tor}(X, f)) \rightarrow H_{n-1}(X)$$

such that the following sequence is exact:

$$\cdots \rightarrow H_n(X) \xrightarrow{f_* - id} H_n(X) \xrightarrow{i_*} H_n(\text{Tor}(X, f)) \xrightarrow{\partial_n} H_{n-1}(X) \rightarrow \cdots$$

◦ **Proof ...**

□

12.10.2 Homology and Cell Attachments

♠ **Proposition 12.10.2.1** ...

12.11 Homology Groups and Direct Limits

12.12 Cellular Homology

◇ **Lemma 12.12.0.1** (Skeleton- H_* -Lemma) Let X be a CW-complex and $n \in \mathbb{N}$,

-
-
-
-

◦ **Proof ...**

□

12.12.1 Definitions

12.13 The Euler Characteristic

Definition 12.13.0.1 Define:

$$\chi(X) := \sum_{n \in \mathbb{N}} (-1)^n$$

12.13.1 The Euler Characteristic of CW-Complex

♠ **Proposition 12.13.1.1** (Homology- χ -Proposition) Given any finite CW-complex X we have

...

◦ **Proof ...**

□

★ Corollary 12.13.1.2 (Homotopy Type χ -Corollart)

◦ **Proof ...**

□

12.13.2 Betti Number

Definition 12.13.2.1 Let (X, A) be a pair:

- Define the **Betti number** of (X, A) to be

$$b_n(X, A) := \text{rank}(H_n(X, A))$$

- If $\sum_{n \in \mathbb{N}} b_n(X, A)$ is finite, then define the **Euler characteristic** of (X, A) to be

$$\chi(X, A) := \sum_n (-1)^n \cdot b_n(X, A)$$

◇ **Lemma 12.13.2.2** (Relative- χ -Lemma)

◦ **Proof ...**

□

12.13.3 Calculation of Euler Characteristics
12.13.4 Applications

Euler's Formula

12.14 The Hurewicz Theorem
12.15 Jordan-Brouwer Separation Theorem
12.16 Moore Space

Definition 12.16.0.1 ...

Chapter 13 Localization

Chapter 14 Note

14.1 4/15

- Review: Homology
- Definition: Cochain, Cohomology, morphisms,
- Cochain homotopy $f, g : C \rightarrow D$

$$s^n : C^n \rightarrow D^{n-1} \quad d^{n-1} \circ s^n + s^{n+1} \circ d^n = f^n - g^n$$

- Functor

$$D : (C^n) \mapsto \text{Coch}(A) \rightarrow Ch(A)$$

♠ Proposition 14.1.0.1

$$H^n(DC) = H_{-n}(C)$$

◦ **Proof ...**

□

♠ Proposition 14.1.0.2 (Commutates with H^n functor) $(H^n f)([x]) = [f^n(x)]$

- Meaning of cochain homotopy: Induces the same map on H^n ;
- Long exact sequence $\rightarrow H_n(A) \rightarrow \dots \rightarrow H_n(C) \xrightarrow{\partial} H^{n+1}(A) \rightarrow \dots, \partial[x] = [y], f^{n+1}$
- Consider $Hom(-, A)$ has an inverse direction of chain complex.

$$d^n : Hom(C, A)^n \rightarrow Hom(C, A)^{n+1}$$

$$f \mapsto f \circ d_{n+1}$$

$Hom(-, A)$ Contravariant functor $Ch(A) \rightarrow Ab$.

•

$$H^n(Y, A) := H^n(Hom(C(Y, \mathbb{Z}), A))$$

Relative $Y' \subseteq Y$

$$H^n(Y, Y', A) := H^n(Hom(\frac{C(Y, \mathbb{Z})}{C(Y', \mathbb{Z})}, A))$$

- X space, $X' \subseteq X$ (especially, $X' = \emptyset$)

$$H^n(X, X', A) := H^n(S(X), S(X'), A)$$

- We have

$$Hom(C_n(Y, \mathbb{Z}), A)^n = \prod_{Y_n} A$$

- $Y' \subseteq Y$ subsimplicial set,

$$Hom\left(\frac{C_*(Y, \mathbb{Z})}{C_*(Y', \mathbb{Z})}, A\right)_n = Hom\left(\frac{C_n(Y, \mathbb{Z})}{C_n(Y', \mathbb{Z})}, A\right) = Hom\left(\frac{\pi(Y_n)}{\pi(Y'_n)}, A\right) = \{f\}$$

Thus, exact

$$0 \rightarrow Hom\left(\frac{C_*(Y, \mathbb{Z})}{C_*(Y', \mathbb{Z})}, A\right) \rightarrow$$

- Homotopy-Invariance

♠ **Proposition 14.1.0.3** If $f, g : X \rightarrow Y$ homotopy chain map, then $H^n(f; A) = H^n(g; A) : H_n(X; A) \rightarrow H_n(Y; A)$.

- ♣ **Example 14.1.0.4** ...

- Warning $Hom(-, A)$ may not generally sends q-i to q-i

$$\dots \quad 0 \quad \mathbb{Z} \quad \mathbb{Z} \quad 0 \quad \dots$$

- ♣ **Example 14.1.0.5**

$$\dots \quad 0 \quad 0 \quad \mathbb{Z}/2 \quad 0 \quad \dots$$

- Review: (Whitehead) For chain complex every quasi-isomorphism $C \xrightarrow{\delta} D$ be dege map free chain complex is a chain homotopy equivalence.
- Consider excoi triple $(U \subseteq Y \subseteq X, \bar{U} \subseteq \mathbb{I}Y)$. $Hom(-, A)$ is a q-i for the admisiible cone $X = (X - U) \cup Y$ the

$$H^n(X - Y, Y - Y, A) \cong H^n(X; Y; A)$$

- $H^m(S^n, A) = \begin{cases} A \oplus A & m = n \\ A & \\ 0 & else \end{cases}, H^m(D^n, S^n, A) =$

- Cellular cochain complex: X CW-Complex, A ab group

Definition 14.1.0.6 $C_{cell}^*(X, A) = Hom(C_*^{cell}(X, \mathbb{Z}), A)$

See: $C_n^{cell}(X, \mathbb{Z}) = H_n(X_n, X_{n-1}, \mathbb{Z})$, $C_{cell}^n(X, A) = H^n(X_n, X_{n-1}, A)$

- Natural isomorphism

$$H^*(C_{cell}^*(X, A)) \cong H^n(X, A)$$

- ♣ **Example 14.1.0.7** $\mathbb{C}P^n$.

Chapter 15 Simplicial Cohomology

15.1 Chirality

Definition 15.1.0.1 ...

15.1.1 Hyperplane-Reflection

◇ **Lemma 15.1.1.1** ...

15.2 Cochain Complexes and Dual Complexes

15.2.1 Preparations

The Hom-Functor

Consider

$$Hom(-, X) \quad f : A \rightarrow B \Rightarrow f^* : Hom(B, G) \rightarrow Hom(A, G)$$

◇ **Lemma 15.2.1.1**

- Given an abelian group G , the map $A \mapsto Hom(A, G), (f : A \rightarrow B) \mapsto (f^* : Hom(B, G) \rightarrow Hom(A, G))$
- If R is a commutative ring, then the above maps

$$A \mapsto Hom(A, R) \quad f \mapsto f^*$$

defines a contravariant functor from the category groups to Mod_R .

◦ **Proof** Omitted. □

◇ **Lemma 15.2.1.2** ...

◦ **Proof** ... □

Cochain Complexes

Cochain Maps

15.2.2 Calculation of Some Cohomology

Empty Space

Path Component H^* -Lemma

title

title

15.2.3 Basic Properties of Cohomology Groups

15.2.4 Reduced Cohomology Groups

15.3 The Cohomology Group of Topological Spaces

15.3.1 Singular Cohomology Groups

Definition 15.3.1.1 Let (X, A) be a pair of topological spaces, and let G be an abelian group.

- For each $n \in \mathbb{N}$, write

$$C^n(X, A; G) := \text{Hom}(C_n(X, A), G) \quad \delta_n := \partial_{n+1}^* : C^n(X, A; G) \rightarrow C^{n+1}(X, A; G)$$

We use the following notations:

- $C^n(X, A; G)$ is called the **singular n -cochain**.
- The kernel of

$$\delta_n : C^n(X, A; G) \rightarrow C^{n+1}(X, A; G)$$

as **singular n -cocycles**.

- The image of

$$\delta_n : C^n(X, A; G) \rightarrow C^{n+1}(X, A; G)$$

as **singular n -coboundaries**.

- Define the cochain complex

$$\dots \leftarrow C^{n+1}(X, A; G) \xleftarrow{\delta_n} C^n(X, A; G) \xleftarrow{\delta_{n-1}} \dots \xleftarrow{\delta_0} C^0(X, A; G) \leftarrow 0$$

the cohomology groups

$$H^n(X, A; G) := \ker(\delta_n) / \text{im}(\delta_{n-1})$$

- Especially, $A = \emptyset$, $C^n(X; G) := C^n(X, \emptyset; G)$.

Morphisms induced by Continuous Maps

Definition 15.3.1.2 Consider a continuous map

$$f : (X, A) \rightarrow (Y, B)$$

Then, the induced map

$$f^* : C^*(Y, B; G) \rightarrow C^*(X, A; G)$$

And the group homomorphism

$$f^* : H^n(Y, B; G) \rightarrow H^n(X, A; G)$$

◇ **Lemma 15.3.1.3** ...

◦ **Proof** ...

□

15.3.2 Example: Explicit Cochains and Cocycles

Circles

Consider the topological space $X = S^1$, with

$$p : \mathbb{R} \rightarrow S^1 \quad t \mapsto \exp(2\pi it)$$

be the universal covering of S^1 . Then, let

$$\sigma^1 : \Delta^1 \rightarrow S^1$$

be singular 1-simplex. Δ^1 is simply connected, by the covering existence theorem

...

◇ **Lemma 15.3.2.1** (θ -Cocycle Lemma)

-
-

title

15.4 Cellular Cohomology and the Kronecker Pairing

15.4.1 Cellular Cohomology

Review: Cellular Cohomology

Definition 15.4.1.1 Let X be a CW-complex:

- Denote X^n the n -skeleton
- Write

$$C_n^{CW}(X) := H_n(X^n, X^{n-1})$$

We refer to the elements of $C_n^{CW}(X)$ as **cellular chain**.

- –
-

- We denoted by

$$d_n : C_n^{CW}(X) \rightarrow C_{n-1}^{CW}(X)$$

the cellular boundary map.

-
-

Definition 15.4.1.2 (Cellular Cohomology) Let X be a CW-complex, G an abelian group:

- Let $n \in \mathbb{N}$:
 - We refer to elements of $\text{Hom}(C_n^{CW}(X), G)$ as **cellular n -cochains**;
 - If X is charted, then we have

$$\text{Hom}(C_n^{CW}, G) = \text{maps from the set of } n\text{-cells to } G = G^{\#n\text{-cells}}$$

- Given any $n \in \mathbb{N}$, we write $\delta_n := d_{n+1}^*$, we refer to

$$\dots \leftarrow \text{Hom}(C_{n+1}^{CW}(X), G) \xleftarrow{\delta_n} \text{Hom}(C_n^{CW}, G) \xleftarrow{\delta_{n-1}} \text{Hom}(C_{n-1}^{CW}, G) \leftarrow \dots$$

The, the n -th cohomology group of X with G coefficients as

$$H_{CW}^n(X; G) := H^n(C_*^{CW}(X), G) = \ker(\delta_n) / \text{im}(\delta_{n-1})$$

Morphisms

Definition 15.4.1.3 ...

◇ **Lemma 15.4.1.4** ...

◦ **Proof** ...

□

♠ **Proposition 15.4.1.5** ...

◦ **Proof** ...

□

Finitely Generated Cohomology

★ **Corollary 15.4.1.6** (Finitely Generated Cohomology Corollary) Let X be a topological space. Suppose we are in one of following two situations:

- X is homotopy equivalent to a retract of a finite CW-complex;
- X is compact topological manifold.

Then, for every $n \in \mathbb{N}$, the cohomology group $H^n(X, \mathbb{Z})$ is a finitely generated group.

◦ **Proof** ...

□

Kronecker Pairing

◇ **Lemma 15.4.1.7** Let C_* be a chain complex and let G be an abelian group:

- – The map

...

–

- – Let $f : C_* \rightarrow D_*$ be a chain map between two chain complexes, and let G be an abelian group. Then, for any $c \in H_n(C)$ and any $\varphi \in H^n(D; G)$ we have:

$$\langle f^*(\varphi), c \rangle = \langle \varphi, f_*(c) \rangle$$

- Let $f : (X, A) \rightarrow (Y, B)$ be a continuous map between two pairs of topological spaces. For any $c \in H_n(X, A)$ and any $\varphi \in H^n(Y, B; G)$ we have:

...

•

◦ **Proof ...**

□

◇ **Lemma 15.4.1.8** (Kronecker Basis Lemma)

◦ **Proof ...**

□

15.4.2 Calculations

Torus

◇ **Lemma 15.4.2.1** (Torus- H^1 -Lemma) We denote by

$$\theta_{\mathbb{Z}} : C_1(S^1) \rightarrow \mathbb{Z}$$

the singular 1-cocycle. By $\mapsto$. Then, two projections

$$p, q : S^1 \times S^1 \rightarrow S^1$$

Then, $p^*([\theta_{\mathbb{Z}}])$ and $q^*([\theta_{\mathbb{Z}}])$ form a basis for

$$H^1(S^1 \times S^1; \mathbb{Z}) \cong \mathbb{Z}^2$$

◦ **Proof ...**

□

Surface- H^1 -Lemma

◇ **Lemma 15.4.2.2** Let $g \in \mathbb{N}_{\geq 2}$. We introduce the following notations

- The surface of genus g , Σ_g ;
- In figure below, we show the projection

...

- Let $\alpha, \beta \in H^1(S^1 \times S^1; \mathbb{Z})$ be the basis taht is given by the torus H^1 -lemma.

With this notation a basis for $H^1(\Sigma_g; \mathbb{Z}) \cong \mathbb{Z}^{2g}$ given by $p_i^*(\alpha), p_i^*(\beta)$

◦ **Proof** ...

□

15.5 Excision and Mayer-Viertoris for Cohomology

15.5.1 Excision in H^* Theorem

♠ **Proposition 15.5.1.1** (Also-Iso-on-Cohomology Proposition)

◦ **Proof** ...

□

► **Theorem** ...

◦ **Proof** ...

□

15.6 The UCT for Cohomology Group

15.7 The Cup Product

Recall:

- Δ the category with objects the finite ordered set

$$[n] := (0 < 1 < \dots < n)$$

With morphisms of order-preserving maps.

A **simplicial set** K is a contravariant functor

...

- The singular complex $S(X)$ of a topological space X

...

15.7.1 First Definition

Tensor Product of Chain Complexes

Recall

$$(C \otimes D)_n := \bigoplus_{p+q=n} (C_p \otimes D_q)$$

$$\partial^\otimes : (C \otimes D)_n \rightarrow (C \otimes D)_{n-1} \quad c_p \otimes d_q \mapsto \partial_p(c_p) \otimes d_q + (-1)^p \cdot c_p \otimes \partial_q(d_q)$$

Definition 15.7.1.1

- (Cycle-Tensor Lemma) The map $\mu : H_p(C) \times H_q(D) \rightarrow H_{p+q}(C \otimes D), [c] \otimes [d] \mapsto [c \otimes d]$ is well-defined homomorphism;
- Given topological space, then from Eilenberg-Zilber theorem we have the natural chain
- **(Cross Product)**

Definition 15.7.1.2 Let X be a topological space, and let R be a commutative ring. We define the **cup product** on the cohomology groups of X as the composition of the following maps

...

15.8 The Relative Cup Product

15.9 The Cap Product

Chapter 16 Orientation

16.1 Orientation of Smooth Manifold

Given in 'Smooth Manifold', here.

16.2 Orientation of Topological Manifolds

16.2.1 Local Homology

Firstly, consider the reflection

$$f_i : \mathbb{R}^n \rightarrow \mathbb{R}^n, (x_1, \dots, x_n) \mapsto (x_1, \dots, x_{i-1}, -x_i, x_{i+1}, \dots, x_n)$$

Now, consider the restriction map to a self-map $S^{n-1} \subseteq \mathbb{R}^n$, acts by -1 on $H_{n-1}(S^{n-1}; \mathbb{Z})$. Isomorphism

$$H_n(\mathbb{R}^n, \mathbb{R}^n - \{0\}; \mathbb{Z}) \xrightarrow{\cong} H_{n-1}(\mathbb{R}^n - \{0\}; \mathbb{Z}) \xleftarrow{\cong} H_{n-1}(S^{n-1}; \mathbb{Z})$$

The first from the relative homology sequence, and the second from $\mathbb{R}^n - \{0\} \cong S^{n-1}$.

Definition 16.2.1.1 Given a pointed topological space (X, x_0) and given a commutative ring R .

The homology group $H_n(X, X - \{x_0\}; R)$ is called the **n -th local homology group with R -coefficients** of X at x_0 .

◇ **Lemma 16.2.1.2** (Manifold-Local R -Homology Lemma) Let M be an n -dimensional topological manifold, and let R be a commutative ring.

For every $k \in \mathbb{N}, x \in M$ we have

- $H_k(M, M - \{x\}) \cong \begin{cases} \mathbb{Z} & x \in M - \partial M, k = n; \\ 0 & \text{Elsewise} \end{cases};$

- The following map is an isomorphism of R -modules

$$H_k(M, M - \{x\}) \otimes R \rightarrow H_k(M, M - \{x\}; R) \quad \sum [\sigma_i] \otimes r_i \mapsto \sum [\sigma_i \otimes r_i]$$

-

◦ **Proof ...**

□

◇ **Lemma 16.2.1.3** (Local- H_* -Reflection Lemma) Let $n \in \mathbb{N}$, and $P \in \mathbb{R}^n$, and

$$\rho : \mathbb{R}^n \rightarrow \mathbb{R}^N$$

◦ **Proof ...**

□

Standard Generators

Definition 16.2.1.4 Let $n \in \mathbb{N}$, we write

$$\partial\Delta^n := \cup_{i=1}^n \{(t_0, \dots, t_n) \in \Delta^n | t_i = 0\}$$

Now, we refer to $\dots \Delta^n := \Delta^n - \partial\Delta^n$ as the interior of Δ^n .

◇ **Lemma 16.2.1.5** (Identity-is-Generator Lemma) For any $k \in \mathbb{N}$, we have

$$H_k(\Delta^n, \partial\Delta^n; R) = \begin{cases} 0 & k \neq n \\ R \cdot [id_{\Delta_n} \otimes 1_R] & k = n \end{cases}$$

◦ **Proof**

- By the original identity is generator lemma,

$$H_k(\Delta^n, \partial\Delta^n) = \begin{cases} \dots \end{cases}$$

•

□

Definition 16.2.1.6 ...

◇ **Lemma 16.2.1.7** (Local H_* -Basics Lemma)

◦ **Proof** ...

□

Orientation Preserving

Definition 16.2.1.8 ...

16.2.2 R -Section on Topological Manifolds

Definition 16.2.2.1 Let M be an n -dimensional topological manifold, and let R be a commutative ring:

-
-
-
-

16.2.3 Orientations of Topological Manifolds

◇ **Lemma 16.2.3.1** ...

◦ **Proof** ...

□

16.2.4 Orientation Covering of Topological Manifold

♠ **Proposition 16.2.4.1** ...

◦ **Proof** ...

□

Definition 16.2.4.2 Let M be a connected topological manifold, $x_0 \in M$.

★ **Corollary 16.2.4.3** ...

◦ **Proof** ...

□

16.2.5 Orientation-Preserving Local Homomorphisms

16.3 Induced Orientation

16.3.1 Induced Orientations

◇ **Lemma 16.3.1.1** (Topological Manifold-induced Orientation Lemma) Let R be a commutative ring and M a topological manifold, that is equipped with an R -orientation:

16.3.2 Glueing Topological Manifolds

Chapter 17 Fibration, Cofibration

17.1 Left, Right Lifting Property

Definition 17.1.0.1 Consider the following diagram: Given fixed $(A, B, X, Y; i : A \rightarrow B; p : X \rightarrow Y)$

$$\begin{array}{ccc} A & \xrightarrow{\forall f} & X \\ \downarrow i & \exists \nearrow & \downarrow p \\ B & \xrightarrow{\forall g} & Y \end{array}$$

If the diagram commutes for all f, g

- i has **left lifting property** p with aspect to a morphism i ;
- p has **right lifting property** i with aspect to a morphism p .

Denoted by

$$i \sqsupseteq p$$

Especially:

- If $A = \emptyset$, and p is surjection, then the diagram is the property of projective module;
- If $Y = \emptyset$, and i is injection, then the diagram is the property of injective module;

Definition 17.1.0.2 (Orthogonal Class/Quillen Negation) Give na class $M \subseteq Mor(C)$. Then, its

- Left weak orthogonal calss/left Quillen negation

$$M^\square := \{p \in Mor(C) | \forall_{i \in M} i \sqsupseteq p\}$$

- Right weak orthogonal class

$$\square M := \{i \in Mor(C) | \forall_{p \in M} i \sqsupseteq p\}$$

17.1.1 Homotopy Lifting Property, Homotopy Lifting Extension Property

HLP

Definition 17.1.1.1 Given: (Y, π) where $\pi : E \rightarrow B$ is a map; Y is a topological space.

We say, the pair satisfies **homotopy lifting property** with respect to Y if:

- For any homotopy $f_\bullet : Y \times I \rightarrow B$;
- For any map $\tilde{f}_0 : Y \rightarrow E$ lifting

$$f_0 = f_\bullet|_{Y \times \{0\}}$$

so that

then, there exists a homotopy $\tilde{f}_\bullet : Y \times I \rightarrow E$ lifting $f_\bullet$, which also satisfies $\tilde{f}_0 = \tilde{f}|_{Y \times \{0\}}$.

$$\begin{array}{ccc}
Y & \xrightarrow{\tilde{f}_0} & E \\
\downarrow i_0 & \nearrow \tilde{f}_\bullet & \downarrow \pi \\
Y \times I & \xrightarrow{f_\bullet} & B
\end{array}$$

Homotopy Lifting Extension Property

Definition 17.1.1.2 Let: $X \supseteq Y$, for simplicity we denote:

$$T : (X \times \{0\}) \cup (Y \times [0, 1]) \subseteq X \times [0, 1]$$

Given additionally a map $\pi : E \rightarrow B$, one says that (X, Y, π) has the **homotopy lifting extension property** if

- For any homotopy, $f : X \times [0, 1] \rightarrow B$;
- For any lifting $\tilde{g} : T \rightarrow E$ of $g = f|_T$, there exists a homotopy

$$\tilde{f} : X \times [0, 1] \rightarrow E$$

which covers f . i.e. $\pi \tilde{f} = f$ and extends $\tilde{g}$ i.e. $\tilde{f}|_T = \tilde{g}$.

$$\begin{array}{ccc}
T & \xrightarrow{\tilde{g}} & E \\
\downarrow i & \nearrow \tilde{f} & \downarrow \pi \\
X \times [0, 1] & \xrightarrow{f} & B
\end{array}$$

Especially, if $Y = \emptyset$, then the property is the HLP.

17.1.2 Hurewicz Fibration/Cofibration

Hurewicz Connections

Definition 17.1.2.1

- (Mapping Cocylinder) Define:

$$\begin{array}{ccc}
Cocyl(\pi) & \longrightarrow & P(B) \\
\downarrow & & \downarrow \\
E & \longrightarrow & B
\end{array}$$

the cocylinder is the pullback of the diagram, where $P(B)$ is the **path space** in Top .

- A **Hurewicz connection** is any continuous section

$$s : Cocyl(\pi) \rightarrow P(E)$$

of the map $\pi_! : P(E) \rightarrow Cocyl(\pi)$ given by

$$\pi_!(u) = (u(0), \pi \circ u)$$

Connection Consider a Hurewicz fibration

$$p : E \rightarrow B$$

With a path $\gamma \in P(B)$. Then, fibres

$$F_{b_0} := p^{-1}(b_0) \quad F_{b_1} := p^{-1}(b_1)$$

Then, define:

$$\Gamma_\gamma(e) := s(e, \gamma)(1) \in F_{b_1} \subseteq B$$

Hurewicz Fibration

Definition 17.1.2.2

- A continuous map $p : E \rightarrow B$ of topological space is called a **Hurewicz fibration** if it satisfies the **right lifting property** against all continuous functions

$$\sigma_0 : X \cong X \times \{0\} \hookrightarrow X \times I$$

including any topological space X as one end of cylinder over it.

$$\begin{array}{ccc} X & \xrightarrow{f} & E \\ \downarrow \sigma_0 & \nearrow \tilde{F} & \downarrow p \\ X \times I & \longrightarrow & B \end{array}$$

► **Theorem** (Hurewicz, 1955) A map is a Hurewicz fibration, precisely if it admits a Hurewicz connection.

◦ **Proof ...**

□

♠ **Proposition 17.1.2.3** The Hurewicz fibration is closed under:

- Composition;
-

◦ **Proof ...**

□

Fibration-LES Theorem

Fibration-Pullback

◇ **Lemma 17.1.2.4** Let $f : A \rightarrow B$ be a continuous map between topological spaces, and let $q : Z \rightarrow B$ be a fibration. Consider the pullback

$$f^*Z := \{(a, z) | a \in A, q(z) = f(a)\} \subseteq A \times Z$$

Then:

- The pullback map

$$\begin{aligned} p : f^*Z &\rightarrow A \\ (a, z) &\mapsto a \end{aligned}$$

is also a fibration.

Furthermore, for any $a \in A$, the fiber $p^{-1}(a)$ is homeomorphic to $q^{-1}(f(a))$.

- The following diagram commutes:

$$\begin{array}{ccc}
 f^* Z & \xrightarrow{(a,z) \mapsto z} & Z \\
 \downarrow p & & \downarrow q \\
 A & \xrightarrow{f} & B
 \end{array}$$

◦ **Proof**

□

17.1.3 Serre Fibration/Cofibration

Definition 17.1.3.1 ...

◇ **Lemma 17.1.3.2** (Non-Serre Fibration) The map

$$p : S^1 \rightarrow [-1, 1]$$

Given by projection, is not a Serre-fibration.

◦ **Proof ...**

□

Fibration-LES Fibration

► **Theorem** (Fibration-LES) Let $p : Y \rightarrow B$ be a Serre fibration, with

The Long Exact Sequence of a Serre Fibration

17.1.4 Dold Fibration/Cofibration

Definition 17.1.4.1 The morphism $f : E \rightarrow B$ of topological spaces is said to have **weak covering homotopy property** for the space Y , if for all squares

...